IntelXtract

OSINT AUTOMATION PLATFORM

AI-Powered OSINT Automation & Intelligence Platform

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AI-Powered OSINT Automation & Intelligence Platform

Final Project Report

Major Project Submission

Academic Year: 2025–2026

Field	Details
Project Title	IntelXtract: AI-Powered OSINT Automation & Intelligence Platform
Subtitle	Desktop Application for Open-Source Intelligence Collection, Correlation, and Reporting
Domain	Cybersecurity · OSINT · Threat Intelligence · Automation
Primary Language	Python 3.13+
GUI Framework	PySide6 (Qt 6)
Database	SQLite (with forward-only migrations v1→v4)
Architecture	Modular Plugin-Based Architecture
Target Platforms	Windows, Linux, macOS
Distribution	PyInstaller Frozen Binaries, Docker Images, PyPI Wheel/Sdist
License	GPL-3.0-or-later
Version	1.0.0 (Tag: `v1.0.0`)
Repository	`https://github.com/yourorg/IntelXtract`

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Date of Submission: [Date]

This report documents the complete software engineering lifecycle of IntelXtract, from requirements through implementation, testing, packaging, and release. All technical claims are substantiated by source code, test results, and documentation in the accompanying repository.

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2. Certificate

This is to certify that the project report titled **"IntelXtract: AI-Powered OSINT Automation & Intelligence Platform"** submitted by **[Student Name]** bearing roll number **[Roll Number]** in partial fulfillment of the requirements for the degree of **[Degree Name]** in **[Department Name]** is a bonafide record of work carried out under my supervision and guidance.

The project demonstrates comprehensive understanding of software engineering principles, cybersecurity fundamentals, and open-source intelligence methodologies. The implementation follows industry-standard practices including modular architecture, automated testing, continuous integration, and responsible disclosure policies.

Signature: _____
Guide Name: [Guide Name]
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Date: [Date]

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3. Acknowledgement

I would like to express my sincere gratitude to my project guide, **[Guide Name]**, for their invaluable guidance, encouragement, and constructive feedback throughout the development of IntelXtract. Their expertise in cybersecurity and software architecture shaped the technical direction of this project.

I am grateful to the faculty of **[Department Name]** for providing the academic foundation in secure software development, network security, and database systems that underpins this work.

Special thanks to the open-source community whose libraries and frameworks made this project possible: the Python Software Foundation, the Qt Project (PySide6), the NetworkX maintainers, the aiohttp team, and countless others.

I acknowledge the use of the following key technologies: - **Python 3.13+** — core language - **PySide6 6.10.3** — GUI framework - **aiohttp 3.14.4** — asynchronous HTTP client - **aiosqlite 0.22.1** — async SQLite driver - **NetworkX 3.7** — graph algorithms for correlation - **Typer 0.27.2** — CLI framework - **Rich 15.0.0** — terminal formatting - **Jinja2 3.1.6** — template engine for reports - **WeasyPrint** — HTML-to-PDF conversion - **Pillow 12.3.0** — image metadata extraction - **pypdf 6.20.0** — PDF metadata extraction - **cryptography 50.0.2** — API key encryption

Finally, I thank my peers and family for their patience and support during the long development cycles.

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4. Abstract

IntelXtract is a desktop application for Open-Source Intelligence (OSINT) automation that consolidates reconnaissance workflows across domains, IP addresses, email addresses, usernames, websites, certificates, and file metadata into a single platform. The system addresses the fragmentation inherent in manual OSINT research — where analysts juggle dozens of browser tabs, command-line tools, and API consoles — by providing a unified engine that orchestrates 19 specialised collectors, correlates findings into an entity-relationship graph, scores risk using a deterministic rule engine, and produces professional reports in five formats.

The architecture follows a modular plugin-based design with strict separation between the scan engine (pure Python, async-first), the persistence layer (SQLite with forward-only migrations), the user interfaces (Typer CLI and PySide6 GUI), and the reporting subsystem (Jinja2 templates with WeasyPrint PDF backend). The engine implements a priority task scheduler, a bounded-concurrency worker pool with per-task timeouts, a shared async TTL cache, and a thread offload pool for blocking operations. Modules communicate through a typed `ModuleContext` that provides HTTP client access, API key resolution, configuration lookup, and memoization primitives.

Key achievements include: - **19 OSINT modules** across 6 supported target types (Domain, IP, URL, Email, Username, File); file-hash (HASH) is classified but has no collector yet - **Correlation engine** building NetworkX MultiDiGraphs with 8 edge types and confidence scoring - **Risk engine** with 22 deterministic rules across 4 categories (TLS, Headers, DNS, Exposure) with category ceilings preventing single-category dominance - **Encrypted API key vault** using Fernet symmetric encryption with OS keyring fallback - **Scheduled scans** supporting interval and 5-field cron with overlap prevention and change detection - **Case management** for grouping scans into investigations with combined reporting - **Plugin SDK** with fail-soft discovery, manifest validation, and engine-parity adapters - **Comprehensive test suite**: 756 tests, 89.54% branch coverage, fully offline integration tests - **Cross-platform packaging**: PyInstaller binaries, Docker images, PyPI wheel/sdist - **CI/CD pipeline**: GitHub Actions with quality gate (ruff, black, mypy, pytest) and release workflow

The project was developed in 29 phases following a documented phase-gate process, each verified by the quality gate (`ruff check . && black --check . && mypy . && pytest`). The final deliverable passes all quality checks, achieves 89.54% branch coverage (exceeding the 80% floor), and produces deterministic, reproducible outputs suitable for professional intelligence workflows.

Keywords: OSINT, reconnaissance, intelligence automation, correlation, risk scoring, plugin architecture, SQLite, asyncio, PySide6, NetworkX

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8. Abbreviations

Abbreviation	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
ASGI	Asynchronous Server Gateway Interface
CA	Certificate Authority
CI/CD	Continuous Integration / Continuous Deployment
CLI	Command-Line Interface
CSV	Comma-Separated Values
CT	Certificate Transparency
CVE	Common Vulnerabilities and Exposures
DB	Database
DFD	Data Flow Diagram
DKIM	DomainKeys Identified Mail
DMARC	Domain-based Message Authentication, Reporting, and Conformance
DNS	Domain Name System
DNSBL	DNS Blocklist
DoH	DNS over HTTPS
E2E	End-to-End
ER	Entity-Relationship
EXIF	Exchangeable Image File Format
FQDN	Fully Qualified Domain Name
GUI	Graphical User Interface
HIBP	Have I Been Pwned
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
I/O	Input/Output
IP	Internet Protocol

Abbreviation	Expansion
JSON	JavaScript Object Notation
Lang	Language
MD	Markdown
MX	Mail Exchange
NFR	Non-Functional Requirement
OS	Operating System
OSINT	Open-Source Intelligence
OOXML	Office Open XML
PDF	Portable Document Format
PII	Personally Identifiable Information
PTR	Pointer Record (reverse DNS)
PyPI	Python Package Index
RDAP	Registration Data Access Protocol
RSS	Really Simple Syndication
SDK	Software Development Kit
SPF	Sender Policy Framework
SQL	Structured Query Language
SQLite	A self-contained, serverless SQL database engine
SSL	Secure Sockets Layer
TTL	Time To Live
TLS	Transport Layer Security
TOC	Table of Contents
URL	Uniform Resource Locator
UUID	Universally Unique Identifier
WHOIS	Who Is (domain registration directory service)
XSS	Cross-Site Scripting

9. Introduction

Open-Source Intelligence (OSINT) is the discipline of collecting and analysing information from publicly available sources to produce actionable intelligence. In modern cybersecurity practice, OSINT underpins attack-surface discovery, due diligence, brand-protection monitoring, incident triage, and threat-actor profiling. The raw material is abundant — DNS records, WHOIS/RDAP registration data, TLS certificates, HTTP response headers, breach corpora, and public news feeds — but it is scattered across dozens of incompatible services, each with its own interface, rate limits, and output schema.

IntelXtract is a desktop application that unifies this fragmented workflow. It accepts a single indicator — a domain, IP address, URL, email address, username, file hash, or local file — classifies its type, selects an appropriate collection plan, executes a set of specialised async collectors, correlates the results into an entity graph, scores the target's risk with a deterministic rule engine, and renders the outcome as a professional report. The same engine is exposed through two front-ends: a scriptable command-line interface (CLI) and a graphical user interface (GUI) built on PySide6.

The name *IntelXtract* encodes the project's purpose: **Intell**igence e**Xtract**ion. The application is deliberately local-first. It stores all data in a local SQLite database, performs no telemetry, requires no user account, and treats network egress as an explicit, auditable, opt-in event. When an AI assistant is enabled, it is clearly labelled as AI-generated and is disabled by default.

This report documents the complete engineering lifecycle of the system: the problem it solves, the architecture that realises it, the modules that implement each capability, the security and privacy posture, the testing evidence, and the packaging and release machinery. Every claim in this report is grounded in the repository's source code, tests, and documentation.

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10. Background and Motivation

10.1 The OSINT Landscape

OSINT practitioners routinely combine tools such as `dig`, `whois`, `openssl s_client`, `curl`, `theHarvester`, certificate-transparency front-ends, and vendor APIs. Each tool answers one narrow question: *What are the A records? Who registered the domain? When does the certificate expire? What security headers are set?* The analytical value, however, lies in the **connections between these answers**. A certificate's subject-alternative-name field may reveal sibling domains; an MX record may point to a mail provider that implies a breach exposure; a favicon hash may link unrelated hosts to the same infrastructure.

Manual correlation is slow, error-prone, and difficult to reproduce. Analysts re-run the same commands, copy-paste into spreadsheets, and lose provenance. Existing automation platforms tend to be cloud-hosted (raising data-sovereignty concerns), closed-source, or expensive per-query.

10.2 Motivation

Three observations motivated IntelXtract:

1. **Fragmentation is the bottleneck, not collection.** Collectors are individually simple; orchestrating them, normalising their output, and preserving provenance is the hard part.
2. **Analysts need reproducible evidence.** A finding that cannot be traced to a timestamped source is not intelligence. Determinism and auditability are therefore first-class requirements.
3. **Privacy is non-negotiable.** Reconnaissance data is sensitive. A tool that silently uploads targets to a third party is unacceptable for many professional contexts.

These observations shaped the project's guiding principles: **unify collection behind a typed module contract**, **correlate and score deterministically**, **persist everything locally with full provenance**, and **make every network egress explicit**.

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11. Problem Statement

The absence of a unified, local, reproducible OSINT workbench creates concrete problems:

- **Inefficiency.** An analyst targeting a single domain may invoke five to ten distinct tools, each requiring separate input, and then manually merge the outputs. This does not scale to investigations covering dozens of indicators.
- **Inconsistency.** Different tools emit different formats. The same fact (for example, a certificate issuer) may be represented in three ways, complicating comparison across targets and scans.
- **Loss of provenance.** Ad-hoc collection leaves no reliable record of *when* a datum was observed or *which* source produced it, undermining defensibility.
- **No correlation.** Isolated facts are rarely actionable. Without a graph linking domains, IPs, certificates, emails, and entities, relationships are invisible.
- **No prioritisation.** A flat list of findings offers no guidance on what matters. Analysts need a defensible, transparent risk score.
- **Privacy exposure.** Cloud tools exfiltrate target data by design.
- **Poor reproducibility.** Manual steps are hard to re-run identically, so change detection ("what changed since yesterday?") is impractical.

Problem statement: *There is a need for a privacy-preserving, local-first, extensible desktop platform that automates the collection of open-source intelligence across multiple indicator types, correlates the resulting findings into an explainable relationship graph, scores risk deterministically, and produces reproducible, professional reports — all while keeping the analyst in full control of data egress.*

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12. Objectives

The project set the following measurable objectives:

1. **Multi-type collection.** Support the principal target types (Domain, IP, URL, Email, Username, File) with specialised collectors using only passive, publicly available sources.
2. **Unified orchestration.** Provide a single async engine that plans, schedules, executes, and aggregates module runs with deterministic ordering.
3. **Correlation.** Build an entity-relationship graph linking findings across modules with typed edges and confidence scoring.
4. **Explainable risk scoring.** Implement a deterministic rule engine with documented weights, category ceilings, and an auditable rule inventory.
5. **Dual interface.** Deliver a fully scriptable CLI and an equivalent graphical interface sharing one engine.
6. **Local persistence with provenance.** Store targets, scans, findings, reports, schedules, plug-ins, keys, logs, and cases in a versioned local database with forward-only migrations.
7. **Reproducible reporting.** Export byte-consistent reports in at least five formats (JSON, CSV, Markdown, HTML, PDF).
8. **Extensibility.** Provide a documented plug-in SDK so third parties can add collectors without modifying core code.
9. **Security and privacy.** Encrypt API keys at rest; disable all AI/telemetry by default; document the threat model.
10. **Quality.** Achieve ≥80% branch coverage with a passing quality gate (`ruff` , `black` , `mypy` , `pytest`) and fully offline integration tests.
11. **Packaged release.** Package via PyInstaller (Linux x86_64 release binaries), Docker, and PyPI, driven by CI/CD, for cross-platform Python (Linux, macOS, Windows).

73.1 Requirement Fulfilment Matrix

Table 73.1 – Requirement Fulfilment Matrix

Objective	Requirement	Fulfilled in	Evidence
1	Multi-type passive collection (6 supported target types; HASH classified)	Ch. 28, 38–46	Tables 38.1, 28.1–28.2; module listings; registry discovery (19 modules)
2	Unified async orchestration	Ch. 27, 29, 30	`core/engine.py`, `core/scheduler.py`, `core/worker_pool.py`
3	Entity correlation graph	Ch. 47–48	8 edge types; NetworkX MultiDiGraph; Figs 47.1–47.2
4	Deterministic explainable risk scoring	Ch. 49–51	22 rules; category ceilings; rating bands
5	Dual CLI and GUI	Ch. 54–56	Typer CLI; PySide6 GUI
6	Local persistence with provenance	Ch. 35–37	Migrations v1–v4; repositories; audit logs
7	Reproducible 5-format reporting	Ch. 52–53	JSON/CSV/MD/HTML/PDF; Jinja2; WeasyPrint fallback
8	Extensible Plugin SDK	Ch. 59	Manifest-gated plugin loader; PluginBase; adapters
9	Security & privacy	Ch. 60, 63–64	Fernet vault; redaction; threat model; privacy statement
10	Quality ≥80%	Ch. 65–66	756 tests, 89.54% branch coverage; quality gate
11	Packaged release (Python cross-platform)	Ch. 68–69	Wheel/sdist, Linux x86_64 binaries, Docker, CI/CD

13. Scope

13.1 In Scope

- **Passive reconnaissance** against publicly available data sources: DNS-over-HTTPS, RDAP/WHOIS, certificate transparency logs, HTTP(S) response inspection, breach-exposure APIs (key-gated), public news RSS, and local file metadata parsing.
- **Eight classified indicator types** (six with active collectors) and a pluggable module framework.
- **Local desktop usage** on Windows, Linux, and macOS, plus headless CLI and Docker execution.
- **Deterministic correlation and risk scoring.**
- **Report generation** and **scan scheduling** with change detection.
- **Case management** for grouping scans, notes, and evidence.
- **An optional, off-by-default AI assistant** that summarises and explains already-collected evidence.

13.2 Out of Scope

- **Active exploitation or intrusive scanning.** The tool performs no exploitation, no brute-forcing, and no vulnerability probing beyond reading public metadata. This is a deliberate ethical boundary.
- **Authenticated or credentialed access** to third-party systems.
- **Real-time streaming** or threat-feed subscription management.
- **Multi-user server deployment.** The architecture is single-user desktop-first; the Docker image provides headless execution, not a multi-tenant service.
- **Autonomous AI decision-making.** The assistant interprets evidence but never triggers scans or changes data on its own.

13.3 Assumptions

- The host has outbound network access to the configured public sources (or the user operates in an offline/cached mode).
- The analyst is authorised to investigate the targets and complies with applicable law and terms of service.
- Python 3.13 or newer is available for source installs; frozen binaries bundle their own runtime.

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14. Literature Survey and Existing Systems

14.1 Academic and Standards Foundations

OSINT methodology draws on several established bodies of work: the intelligence cycle (planning, collection, processing, analysis, dissemination), MITRE's ATT&CK framework for adversarial technique taxonomy, and standards such as RDAP (RFC 7480–7484) and Certificate Transparency (RFC 6962). These inform IntelXtract's separation of collection from analysis and its emphasis on provenance.

14.2 Existing Tools

Tool	Category	Strengths	Limitations relative to IntelXtract
theHarvester	CLI recon	Broad source coverage, mature	No correlation graph, no risk scoring, no GUI, no persistence/reporting model
Maltego	Graph analysis	Excellent link analysis, transforms	Commercial, GUI-centric, transforms often cloud-hosted, not local-first
SpiderFoot	Automation platform	Many modules, correlator, web UI	Server-oriented, heavier deployment, weaker local desktop story
Recon-ng	CLI framework	Scriptable, modular	Steep learning curve, no first-class risk scoring or report templates
`dig` / `whois` / `openssl`	Point tools	Ubiquitous, precise	No aggregation, no persistence, no correlation, manual re-runs
Shodan/Censys APIs	Data services	Rich internet-wide data	Proprietary, per-query cost, data leaves the analyst's machine
Amass	Attack-surface mapping	Strong DNS/CT enumeration	Focused on enumeration, not scoring/reporting or GUI

14.3 Gap Analysis

The survey reveals a consistent gap: **no single local-first desktop tool combines multi-type passive collection, explainable correlation, deterministic risk scoring, reproducible multi-format reporting, a dual CLI/GUI, and a documented plug-in SDK under a privacy-preserving default posture.** IntelXtract targets this gap directly. Chapter 70 returns to this comparison with a feature-by-feature matrix.

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15. Proposed System

IntelXtract is organised around a small number of cooperating subsystems:

- **Input Engine** — classifies raw indicators into one of eight classified target types (Domain, IP, URL, Email, Username, Hash, File, Unknown) and selects a scan profile (quick or deep) that determines which modules run.
- **Scan Engine** — plans module runs as prioritised tasks, executes them through a bounded-concurrency worker pool with per-task timeouts, and aggregates findings and run metadata into a `ScanResult`.
- **Module Layer** — 19 collectors, each subclassing a common `BaseModule`, each declaring the target types it supports and the fields it produces.
- **Correlation Engine** — constructs a NetworkX `MultiDiGraph` from findings, adding typed edges (`resolves_to`, `registered_by`, `issued_to`, `same_email`, `part_of`, `mentions`, `profile`, `uses`) with confidence weights.
- **Risk Engine** — applies 22 deterministic rules across four categories, applies category ceilings, and derives a 0–100 score and rating.
- **Persistence Layer** — typed repositories over SQLite with forward-only migrations, storing targets, scans, findings, reports, schedules, cases, plug-ins, keys, settings, and logs.
- **Reporting Subsystem** — a shared `ReportModel` consumed by JSON, CSV, Markdown, HTML, and PDF exporters so all formats remain consistent.
- **Interfaces** — a Typer CLI and a PySide6 GUI that share the same engine and repository layer.
- **Assistant** — an optional, off-by-default facade over any OpenAI-compatible endpoint for grounded summarisation and explanation.
- **Plugin SDK** — a manifest-based loader with fail-soft discovery so third-party collectors integrate without core changes.

Figure 15.1 – High-Level Feature Map of IntelXtract

```
mindmap
  root((IntelXtract))
    Collection
      19 modules
      7 target types
      Passive sources
      Key-gated APIs
    Analysis
      Correlation graph
      22 risk rules
      4 categories
      Entity model
    Persistence
      SQLite v4
      Repositories
      Migrations
      Provenance
    Reporting
      JSON
      CSV
      Markdown
      HTML
      PDF
    Interfaces
      CLI
      GUI
      AI assistant
      Plugins
    Operations
      Scheduling
      Change detection
      Cases
      Vault
      CI/CD
```

---

16. Feasibility Study

16.1 Technical Feasibility

The project is built entirely on mature, permissively licensed open-source components (Python, PySide6, aiohttp, aiosqlite, NetworkX, Jinja2, cryptography). No component requires custom infrastructure. The async-first engine relies on language features available since Python 3.5 and is validated on Python 3.13/3.14. Technical feasibility is therefore **high**.

16.2 Economic Feasibility

The stack is free. Development used a single workstation; CI runs on free GitHub Actions runners. Optional key-gated sources (for example, a breach-exposure API) have free tiers. There is no recurring infrastructure cost. Economic feasibility is **high**.

16.3 Operational Feasibility

The tool installs from a wheel, a frozen binary, or Docker, and requires no server, no account, and no database administration. The CLI is scriptable; the GUI is discoverable. Operational feasibility is **high**.

16.4 Legal and Ethical Feasibility

Only passive collection from public sources is performed. The project ships a `SECURITY.md`, an ethical-usage stance in the README, and a privacy document. No intrusive techniques are implemented. Users are responsible for authorisation and lawful use. Feasibility is **acceptable with documented user responsibility**.

16.5 Schedule Feasibility

The work was divided into 29 gated phases (Chapter 19), each producing a verifiable increment. All phases completed. Schedule feasibility was **demonstrated in practice**.

Table 16.1 – Feasibility Analysis Summary

Dimension	Verdict	Key Evidence
Technical	High	Mature OSS stack; validated on Python 3.13/3.14
Economic	High	Zero licence/infra cost; free CI tiers
Operational	High	Wheel/binary/Docker install; no server, no account
Legal/Ethical	Acceptable	Passive-only; documented responsibility
Schedule	Demonstrated	29 gated phases completed

---

17. Functional Requirements

The functional requirements below use the identifier scheme `FR-n` and were traced from the phase plan and verified against tests.

Table 17.1 – Functional Requirements

ID	Requirement	Priority	Verified By
FR-1	Classify a raw indicator into a target type	Must	`tests/core/test_input_engine.py`
FR-2	Select quick/deep scan profile per target type	Must	engine tests
FR-3	Execute modules concurrently with bounded workers	Must	worker-pool tests
FR-4	Enforce per-task timeouts and record TIMEOUT status	Must	engine tests
FR-5	Provide a shared async TTL cache with single-flight	Should	cache tests
FR-6	Offload blocking work to a thread pool	Should	offload tests
FR-7	Deduplicate findings by content hash	Must	engine tests
FR-8	Build a correlation graph with typed, weighted edges	Must	correlation tests
FR-9	Score risk deterministically with rules and ceilings	Must	risk tests
FR-10	Persist targets, scans, findings, reports	Must	repository tests
FR-11	Run forward-only migrations v1→v4	Must	migration tests
FR-12	Export JSON/CSV/Markdown/HTML/PDF reports	Must	exporter tests
FR-13	Provide a full CLI command set	Must	CLI tests
FR-14	Provide a GUI equivalent to the CLI	Should	pytest-qt tests
FR-15	Encrypt API keys at rest	Must	vault tests
FR-16	Support interval and cron schedules	Must	scheduling tests
FR-17	Detect changes between successive scans	Must	diff tests
FR-18	Group scans into cases with notes/evidence	Should	case tests
FR-19	Load third-party plug-ins fail-soft	Should	plugin tests
FR-20	Provide an optional, off-by-default AI assistant	Could	assistant tests
FR-21	Store runtime settings in the DB	Should	settings tests
FR-22	Log redacted audit events	Should	logger tests
FR-23	Report application/dependency health	Should	doctor tests
FR-24	Package as wheel, frozen binaries, Docker	Should	build scripts

18. Non-Functional Requirements

Table 18.1 – Non-Functional Requirements

ID	Requirement	Target	Evidence
NFR-1	Test coverage (branch)	≥ 80%	89.54% measured (Ch. 66)
NFR-2	Determinism	Byte-identical report given identical findings	golden report test
NFR-3	Local-first privacy	No egress unless configured	assistant disabled default; privacy doc
NFR-4	Secret protection	Keys encrypted at rest	Fernet vault (Ch. 60)
NFR-5	Portability	Windows/Linux/macOS	packaging (Ch. 68)
NFR-6	Performance	Engine ≥ thousands of scans/s in-memory	benchmarks (Ch. 67)
NFR-7	Maintainability	Lint + format + types clean	quality gate (Ch. 19)
NFR-8	Extensibility	Add collector without core edits	Plug-in SDK (Ch. 59)
NFR-9	Offline testability	Integration tests need no internet	loopback mock (Ch. 65)
NFR-10	Startup simplicity	Single-command CLI/GUI launch	entry points (Ch. 68)
NFR-11	Robustness	Fail-soft modules; scan never aborts on one failure	per-module error capture
NFR-12	Audability	Every finding has module, timestamp, evidence	provenance model

19. Development Methodology

Development followed an iterative, **phase-gate** methodology. The project was decomposed into 29 phases (numbered 0 through 29), each with a defined deliverable and an explicit acceptance gate. A phase was considered complete only after the full quality gate passed.

Figure 19.1 – Phase-Gate Development Lifecycle

```
flowchart LR
    A[Plan phase] --> B[Implement]
    B --> C{Run quality gate}
    C -->|ruff| D[Fix]
    C -->|black| D
    C -->|mypy| D
    C -->|pytest| D
    D --> B
    C -->|all pass| E[Commit locally]
    E --> F[Next phase]
```

Figure 19.2 – Quality Gate Pipeline

```
flowchart TD
    S[Source] --> R[ruff check .]
    S --> B[black --check .]
    S --> M[mypy .]
    S --> P[pytest -rN]
    R --> G{All green?}
    B --> G
    M --> G
    P --> G
    G -->|yes| C[Commit]
    G -->|no| X[Stop and fix]
```

The gate command is:

```
export PATH="$PWD/.venv/bin:$PATH" && \
  black . && ruff check . && mypy . && \
  pytest -p no:cacheprovider -rN
```

Table 19.1 – Development Phase Summary

Phase	Theme	Representative Deliverable
0	Scaffolding	Package layout, config, logging
1–3	Core foundations	Models, entities, HTTP client, cache
4–6	First modules	Domain/IP collectors, registry
7–9	Engine	Input engine, worker pool, scan engine
10–12	Persistence	SQLite schema, repositories, migrations
13–15	More modules	Email, username, website
16	Correlation	NetworkX graph engine
17	Risk	Rule-based scoring engine
18	Reporting	Shared model + exporters
19–20	CLI	Full command surface
21	GUI	PySide6 application
22	Cases	Investigation grouping
23	Assistant	Optional grounded AI
24	Vault	Encrypted API keys, settings
25	Performance	Pooling, caching, offload, benchmarks
26	Testing	Coverage gate, offline E2E
27	Documentation	Architecture + guides
28	Packaging	PyInstaller, Docker, PyPI
29	Release	v1.0.0, release workflow, tag

> **Note on the phase plan document.** The summary table in `docs/Phase_Plan.md` still shows Phases 18 and 21 as unchecked although they are complete. This is a documentation inconsistency, not an implementation gap; it is recorded here for transparency.

---

20. Use Case Analysis

20.1 Actors

Table 20.1 – Actors and Responsibilities

Actor	Description	Primary Use Cases
Analyst	Human investigator	Scan, review, report, schedule, manage cases
CLI User	Automation/scripted consumer	Batch scans, CI integration, report generation
GUI User	Desktop consumer	Interactive scan, visual exploration
Plugin Author	Third-party developer	Write and register collectors
Scheduler	Internal time-driven actor	Trigger recurring scans
AI Provider	Optional external service	Answer grounded prompts (opt-in)
Administrator	Operator	Configure keys, settings, health checks

20.2 Use Case Diagram

Figure 20.1 – Use Case Diagram

```
flowchart LR
    Analyst((Analyst))
    CLIUser((CLI User))
    Scheduler((Scheduler))
    PluginAuthor((Plugin Author))

    Analyst --> UC1[Perform scan]
    Analyst --> UC2[Review findings]
    Analyst --> UC3[Generate report]
    Analyst --> UC4[Manage case]
    Analyst --> UC5[Schedule scan]
    Analyst --> UC6[Ask assistant]
    CLIUser --> UC1
    CLIUser --> UC3
    CLIUser --> UC7[Batch/CI scan]
    Scheduler --> UC5
    PluginAuthor --> UC8[Add collector]

    UC1 --> UC9[Classify target]
    UC1 --> UC10[Run modules]
    UC1 --> UC11[Persist results]
    UC2 --> UC12[Correlate]
    UC2 --> UC13[Score risk]
    UC3 --> UC14[Export format]
```

20.3 Representative Scenarios

1. **Ad-hoc domain check.** An analyst scans `example.com` in quick mode, reviews the risk rating and top findings, and exports an HTML report.
2. **Deep investigation.** The analyst runs a deep scan, explores the correlation graph, and attaches related findings to a case.
3. **Continuous monitoring.** The analyst schedules a daily scan; the scheduler detects that a new subdomain appeared and records a change event.
4. **Scripted batch.** A CI pipeline runs the CLI against an inventory list and fails if any target scores critical.
5. **Extension.** A plugin author adds a collector for a new data source and enables it from the CLI or GUI.

---

21. System Workflows

21.1 End-to-End Scan Workflow

Figure 21.1 – End-to-End Scan Workflow (Activity Diagram)

```
flowchart TD
    Start([Start]) --> Input[Accept raw indicator]
    Input --> Classify{Classify type}
    Classify -->|unknown| Reject[Reject with error]
    Classify -->|known| Profile[Select quick/deep profile]
    Profile --> Plan[Plan module tasks]
    Plan --> Run[Execute via worker pool]
    Run --> Each{Per-module outcome}
    Each -->|success| Collect[Collect findings]
    Each -->|error/timeout| Record[Record error/timeout]
    Collect --> Dedupe[Deduplicate by content hash]
    Record --> Dedupe
    Dedupe --> Aggregate[Aggregate ScanResult]
    Aggregate --> Persist[(Persist to SQLite)]
    Persist --> Correlate[Build correlation graph]
    Correlate --> Risk[Score risk]
    Risk --> Report[Render report]
    Report --> End([End])
```

21.2 Quick versus Deep Scan

The input engine defines, per target type, a *quick* profile (fast, minimal network) and a *deep* profile (comprehensive, more sources). For a domain, quick runs `dns` and `http`; deep adds `whois`, `ssl`, `subdomain`, `website`, `certificate`, and `news`. For an IP, quick runs `geo` and `rdns`; deep adds `reputation`.

Figure 21.2 – Quick Scan versus Deep Scan (Sequence Diagram)

```
sequenceDiagram
    participant U as User
    participant C as CLI/GUI
    participant E as Engine
    participant M as Modules
    participant D as Database

    U->>C: scan target --mode quick
    C->>E: run(target, quick)
    E->>M: dns, http
    M-->>E: findings
    E-->>C: ScanResult
    C->>D: persist
    C-->>U: summary

    U->>C: scan target --mode deep
    C->>E: run(target, deep)
    E->>M: dns, http, whois, ssl,<br/>subdomain, website, certificate, news
    M-->>E: findings
    E-->>C: ScanResult
    C->>D: persist
    C-->>U: summary
```

---

22. System Architecture Overview

IntelXtract follows a **layered, ports-and-adapters** architecture. A pure-Python asynchronous core contains all collection, correlation, and scoring logic and is completely independent of both the user interfaces and the database. The interfaces and the persistence layer attach to the core through explicit seams (`result_sink` , `classifier` , repository protocols), which keeps the core unit-testable and reusable.

22.1 Layered Architecture

Figure 22.1 – Layered System Architecture

```
flowchart TD
    subgraph Presentation
        CLI[Typer CLI]
        GUI[PySide6 GUI]
    end
    subgraph Application
        Bridge[EngineBridge / asyncio loop]
        Cmd[CLI command handlers]
    end
    subgraph Core
        Input[Input Engine]
        Engine[Scan Engine]
        Sched[Task Scheduler]
        Pool[Worker Pool]
        Corr[Correlation]
        Risk[Risk Engine]
        HTTP[HTTP Client]
        Cache[Async TTL Cache]
        Off[Offload Pool]
        Plug[Plugin Registry]
        Assist[Assistant]
    end
    subgraph Modules
        Collectors[19 OSINT Modules]
        Plugins[Third-party Plugins]
    end
    subgraph Data
        Repos[Repositories]
        DB[(SQLite v4)]
        Vault[Fernet Vault]
        Reports[Report Builder/Exporters]
        Templates[Jinja2 Templates]
    end

    CLI --> Cmd --> Engine
    GUI --> Bridge --> Engine
    Engine --> Input
    Engine --> Sched --> Pool
    Engine --> HTTP
    Engine --> Cache
    Engine --> Off
    Engine --> Corr
    Engine --> Risk
    Engine --> Collectors
    Plug --> Plugins
    Engine --> Plug
    Engine -->|result_sink| Repos
```

```
    Repos --> DB
    Vault --> Repos
    Reports --> Templates
    Reports --> Repos
    Cmd --> Reports
    Bridge --> Reports
    Assist --> Reports
```

The dependency direction is strictly downward: the presentation layer may call the core, but the core never imports the GUI, the CLI, or the database. This is enforced by convention, reviewed per phase, and reflected in the coverage configuration, which measures `core`, `modules`, `reports`, and `plugins` only.

22.2 Deployment Architecture

Figure 22.2 – Deployment Architecture

```
flowchart LR
    subgraph Host["Analyst Workstation (Windows / Linux / macOS)"]
        Binary[IntelXtract GUI / CLI]
        LocalDB[(SQLite intelxtract.db)]
        KeyStore[OS Keyring or 0600 key file]
        Logs[Rotating log file]
    end
    subgraph Optional["Optional / Opt-in Egress"]
        Sources[Public OSINT endpoints]
        Provider[OpenAI-compatible API]
    end
    Binary --- LocalDB
    Binary --- KeyStore
    Binary --- Logs
    Binary -->|passive queries| Sources
    Binary -.->|only if assistant.enabled=true| Provider
```

All persistent state lives on the analyst's machine. Network egress to OSINT sources occurs only when a scan runs, and egress to an AI provider occurs only when the assistant is explicitly enabled. There is no server component and no telemetry endpoint.

---

23. Component Architecture

Figure 23.1 – Component Diagram

```
flowchart TB
    subgraph Engine
        SE[ScanEngine]
        MC[ModuleContext]
        SR[ScanResult]
    end
    subgraph Orchestration
        IE[input_engine.classify/select_modules]
        TS[TaskScheduler]
        WP[WorkerPool]
        OP[OffloadPool]
    end
    subgraph Infrastructure
        CFG[Config]
        LOG[logger]
        HC[HttpClient]
        CA[AsyncTTLCache]
    end
    subgraph Analysis
        ENT[entities]
        COR[correlation]
        RR[risk_rules]
        RKG[risk]
    end
    subgraph Persistence
        REP[repositories]
        PER[persistence]
        MIG[migrations]
    end
    subgraph Reporting
        RB[reports.builder]
        RX[reports.exporter]
    end
    SE --> IE
    SE --> TS
    SE --> WP
    SE --> MC
    SE --> SR
    MC --> HC
    MC --> CA
    MC --> OP
    MC --> CFG
    SE --> ENT
    SE --> COR
    SE --> RKG
    RKG --> RR
    SE -->|result_sink| PER
```

```
    PER --> REP
    MIG --> REP
    RB --> COR
    RB --> RKG
    RX --> RB
```

Table 23.1 – Component Responsibilities

Component	Module	Responsibility
Scan Engine	`core/engine.py`	Orchestrates plan → schedule → execute → aggregate
Module Context	`core/engine.py`	Per-scan services handed to modules (HTTP, cache, keys, offload)
Input Engine	`core/input_engine.py`	Classify, normalise, and profile targets
Task Scheduler	`core/scheduler.py`	Priority queue and per-task state accounting
Worker Pool	`core/worker_pool.py`	Bounded-concurrency execution with timeouts
Offload Pool	`core/offload.py`	Run blocking callables off the event loop
HTTP Client	`core/http_client.py`	Retries, backoff, throttling, conditional caching
Cache	`core/cache.py`	LRU + per-key TTL, single-event-loop
Config	`core/config.py`	Layered defaults → JSON → env
Logger	`core/logger.py`	Redacting console + rotating file, scan context
Entities	`core/entities.py`	Canonical entity extraction and identity
Correlation	`core/correlation.py`	NetworkX entity graph with typed edges
Risk Rules	`core/risk_rules.py`	22 deterministic rule definitions
Risk Engine	`core/risk.py`	Category ceilings, score, rating, summary
Repositories	`database/repositories.py`	Typed CRUD over SQLite
Persistence	`database/persistence.py`	Engine `result_sink`, history, case payloads
Migrations	`database/migrations.py`	Forward-only v1→v4 schema evolution
Report Builder	`reports/builder.py`	Assemble the shared `ReportModel`
Report Exporter	`reports/exporter.py`	JSON/CSV/Markdown/HTML/PDF rendering
Plugin Loader	`core/plugin_loader.py`	Fail-soft discovery and engine adapters

23.1 Package Dependency Graph

Figure 23.2 – Package Dependency Graph

```
flowchart TD
    cli --> core
    cli --> database
    cli --> reports
    cli --> assistant
    gui --> core
    gui --> database
    gui --> reports
    gui --> assistant
    database --> core
    reports --> core
    assistant --> core
    assistant --> reports
    modules --> core
    plugins --> core
    core --> core
```

The graph confirms that `core` is the only widely-shared dependency and that no cycle returns from `core` to the outer layers.

---

24. Data Flow Diagrams

24.1 Level-0 (Context) Diagram

Figure 24.1 – Level-0 Data Flow Diagram (Context)

```
flowchart LR
    Analyst[Analyst / CLI / GUI]
    IX((IntelXtract))
    Sources[Public OSINT Sources]
    Provider[AI Provider opt-in]

    Analyst -->|indicator, mode, options| IX
    IX -->|scan summary, reports| Analyst
    IX -->|passive queries| Sources
    Sources -->|public facts| IX
    IX -.->|grounded prompt, opt-in| Provider
    Provider -.->|AI text, opt-in| IX
```

24.2 Level-1 Diagram

Figure 24.2 – Level-1 Data Flow Diagram

```
flowchart TD
    U[User] -->|raw indicator| P1[1.0 Classify & Normalise]
    P1 -->|type, profile| P2[2.0 Plan Modules]
    P2 -->|task list| P3[3.0 Execute Modules]
    P3 -->|HTTP requests| D1[(External Sources)]
    D1 -->|responses| P3
    P3 -->|raw output| P4[4.0 Normalise & Dedupe]
    P4 -->|findings| P5[5.0 Correlate]
    P4 -->|findings| P6[6.0 Score Risk]
    P5 -->|graph| D2[(SQLite)]
    P6 -->|risk| D2
    P4 -->|findings| D2
    P5 --> P7[7.0 Build Report]
    P6 --> P7
    D2 --> P7
    P7 -->|HTML/PDF/JSON/CSV/MD| U
```

The level-1 diagram shows the seven core processes and two data stores (the SQLite database and the external sources). Provenance travels with each datum: findings retain their module, timestamp, and evidence through every stage.

25. Project Directory Structure

Figure 25.1 – Repository Folder Structure

```
flowchart TD
    Root[IntelXtract/] --> app[app/ - application config]
    Root --> assets[assets/ - icons, brand]
    Root --> assistant[assistant/ - optional AI facade]
    Root --> cli[cli/ - Typer CLI]
    Root --> config[config/ - settings.json]
    Root --> core[core/ - engine, analysis, infra]
    Root --> database[database/ - schema, repos, migrations]
    Root --> docs[docs/ - guides and design]
    Root --> examples[examples/ - sample reports]
    Root --> gui[gui/ - PySide6 UI]
    Root --> modules[modules/ - 19 collectors]
    Root --> plugins[plugins/ - third-party]
    Root --> reports[reports/ - builder + exporters]
    Root --> scripts[scripts/ - build, bench, pyinstaller]
    Root --> templates[templates/ - Jinja2]
    Root --> tests[tests/ - pytest suites]
```

The `modules/` package is organised by target family:

- `modules/domain/` — `dns`, `http`, `ssl`, `subdomain`, `whois`
- `modules/ip/` — `geo`, `rdns`, `reputation`
- `modules/website/` — `favicon`, `headers`, `http_methods`, `robots`, `tech`
- `modules/email/` — `email`, `breach`
- `modules/username/` — `username`
- `modules/certificate/` — `certificate`
- `modules/metadata/` — `metadata`
- `modules/news/` — `news`

---

26. Technology Stack

Table 26.1 – Technology Stack

Layer	Technology	Rationale
Language	Python 3.13+	Rich async ecosystem, cross-platform
Async runtime	`asyncio`	Concurrency for I/O-bound collection
GUI	PySide6 (Qt 6)	Native look, mature widgets, graphics
CLI	Typer + Rich	Typed commands, styled terminal output
HTTP	aiohttp	Async client with connection pooling
Database	SQLite via aiosqlite	Serverless, local-first, async
Graph	NetworkX	MultiDiGraph algorithms for correlation
Templates	Jinja2	HTML report rendering
PDF	WeasyPrint (optional)	HTML→PDF with graceful fallback
Crypto	`cryptography` (Fernet)	Authenticated symmetric encryption
Metadata	Pillow, pypdf	Image EXIF and PDF metadata
Testing	pytest, pytest-asyncio, pytest-cov, pytest-qt	Async, coverage, Qt UI tests
Quality	ruff, black, mypy	Lint, format, static typing
Packaging	PyInstaller, build, Docker	Binaries, wheel/sdist, containers
CI/CD	GitHub Actions	Quality gate and release automation

Table 26.2 – Runtime Dependency Versions

Package	Version	Purpose
aiohttp	3.14.4	Async HTTP
aiosqlite	0.22.1	Async SQLite driver
typer	0.27.2	CLI framework
rich	15.0.0	Terminal formatting
jinja2	3.1.6	Templating
PySide6	6.10.3	GUI
cryptography	50.0.2	Encryption
networkx	3.7	Graph analysis
Pillow	12.3.0	Image parsing
pypdf	6.20.0	PDF parsing

Optional: `weasyprint` (PDF; if absent, a print-friendly HTML fallback is produced with a warning), `keyring` (OS keyring vault backend).

---

27. Core Engine Architecture

The scan engine is the heart of IntelXtract. `ScanEngine.scan()` implements a fixed pipeline: validate → classify → plan → schedule → execute → aggregate → correlate → score → sink.

Figure 27.1 – Core Engine Class Diagram

```
classDiagram
    class ScanEngine {
        +register(module)
        +module_names() list
        +scan(target, mode, module_names) ScanResult
        -_validate_target(target) str
        -_plan(...) tuple
        -_build_result(...) ScanResult
    }
    class ModuleContext {
        +config
        +cache
        +http
        +offload
        +api_key(name) str
        +cached(key, factory, ttl)
        +to_thread(func, ...)
    }
    class ScanResult {
        +scan_id
        +target
        +target_type
        +mode
        +status
        +runs list
        +findings list
        +correlation dict
        +risk dict
    }
    class ModuleRun {
        +module
        +status
        +duration
        +result
        +error
    }
    class TaskScheduler {
        +submit(coro, ...) Task
        +next_task() Task
        +counts() dict
    }
    class WorkerPool {
        +run() PoolStats
        +cancel()
    }
    ScanEngine --> ModuleContext
    ScanEngine --> ScanResult
```

```
    ScanResult --> ModuleRun
    ScanEngine --> TaskScheduler
    ScanEngine --> WorkerPool
```

Table 27.1 – Core Engine Components

Component	Role
`ScanEngine`	Public facade; owns config, cache, classifier, sink
`ModuleContext`	Service container passed to each `module.run`
`ScanResult`	Immutable-ish structured outcome with runs, findings, graph, risk
`ModuleRun`	Per-module status/duration/result/error
`Task` / `TaskScheduler`	Prioritised task queue with state accounting
`WorkerPool`	Consumes tasks with bounded concurrency and timeouts

27.1 Scan Sequence

Figure 27.2 – Scan Sequence Diagram

```
sequenceDiagram
    participant Caller
    participant Engine as ScanEngine
    participant Input as input_engine
    participant Sched as TaskScheduler
    participant Pool as WorkerPool
    participant Mod as Module
    participant Corr as correlation
    participant Risk as risk
    participant Sink as result_sink

    Caller->>Engine: scan(target, mode)
    Engine->>Engine: validate target
    Engine->>Input: classify(target)
    Input-->>Engine: TargetType
    Engine->>Input: select_modules(type, mode)
    Input-->>Engine: module names
    loop each planned module
        Engine->>Sched: submit(module.run(...), timeout)
    end
    Engine->>Pool: run()
    Pool->>Sched: next_task()
    Sched-->>Pool: Task
    Pool->>Mod: await run(target, ctx)
    Mod-->>Pool: ModuleResult
    Pool->>Sched: mark_done(task, result)
    Pool-->>Engine: PoolStats
    Engine->>Engine: build ScanResult (dedupe findings)
    Engine->>Corr: build_graph(findings, target, type)
    Corr-->>Engine: graph / None
    Engine->>Risk: assess_risk(findings, target)
    Risk-->>Engine: RiskReport
    Engine->>Sink: await sink(result)
    Engine-->>Caller: ScanResult
```

Robustness guarantee: module failures never abort a scan. Exceptions raised inside a module are captured per task and reported as `FAILED` or `TIMEOUT` runs; correlation and risk assessment are individually guarded so that an analysis error degrades gracefully to `None` rather than failing the scan.

---

28. Input Engine and Target Classification

The input engine converts a raw string into a canonical target and a collection plan. Classification uses an ordered precedence (first match wins), documented in the module docstring and reproduced here.

Figure 28.1 – Target Classification Flowchart

```
flowchart TD
    S([Raw input]) --> E{empty?}
    E -->|yes| U[UNKNOWN]
    E -->|no| URL{schema or host/path?}
    URL -->|yes| TURL[URL]
    URL -->|no| EMAIL{user@host valid?}
    EMAIL -->|yes| TEMAIL[EMAIL]
    EMAIL -->|no| IP{IPv4/IPv6?}
    IP -->|yes| TIP[IP]
    IP -->|no| HASH{32/40/64 hex?}
    HASH -->|yes| THASH[HASH]
    HASH -->|no| FILE{path or known ext?}
    FILE -->|yes| TFILE[FILE]
    FILE -->|no| DOMAIN{hostname with dot?}
    DOMAIN -->|yes| TDOMAIN[DOMAIN]
    DOMAIN -->|no| USER{handle pattern?}
    USER -->|yes| TUSER[USERNAME]
    USER -->|no| U
```

Table 28.1 – Target Types and Classifiers

Type	Recognition rule	Normalisation
URL	`scheme://…` or host-with-path/path	Default scheme `https`, punycode host, lowercase scheme
EMAIL	`user@host` with valid host	Lowercase host, preserve local part
IP	IPv4/IPv6 (port via `[v6]:port`)	Compressed canonical form
HASH	32/40/64 lowercase hex	Lowercase
FILE	Path separators, drive letter, `~`, dotfile, known extension	Verbatim
DOMAIN	Valid hostname with ≥1 dot	Punycode, lowercase, strip trailing dot
USERNAME	Bare handle `[A-Za-z0-9][A-Za-z0-9_.-]{1,31}`	Strip leading `@`
UNKNOWN	Nothing matched	Verbatim

Documented trade-offs: `example.md` classifies as FILE (the file-extension rule precedes the domain rule); `john.doe` classifies as DOMAIN; `localhost:3000` classifies as USERNAME; IPv6 with a port requires bracket notation.

28.1 Scan Profiles

Table 28.2 – Engine Scan Profiles

Target type	Quick modules	Deep modules
Domain	dns, http	whois, dns, ssl, subdomain, http, certificate, news (+ website)
IP	geo, rdns	geo, rdns, reputation
URL	headers, http_methods	headers, tech, favicon, robots, http_methods, certificate
Email	email	email, breach, domain
Username	username	username
File	metadata	metadata
Hash	—	—

The deep profile is also used for `custom` scans; a caller may pass explicit module names, which are validated against the registry. Unknown module names raise a `ValidationError` before any execution begins.

---

29. Task Scheduling and Prioritization

`TaskScheduler` is a priority queue backed by `asyncio.PriorityQueue`. Lower priority numbers run first; equal priorities run in FIFO order via a monotonically increasing sequence counter. Worker sentinels are inserted at the lowest possible priority so they are dequeued only after all real work.

Figure 29.1 – Scheduling Architecture

```
flowchart LR
    P[Planner] -->|submit coro, name, timeout| Q[(PriorityQueue)]
    Q -->|next_task| W1[Worker 1]
    Q -->|next_task| W2[Worker 2]
    Q -->|next_task| WN[Worker N]
    W1 --> ST[mark_running/done/failed/cancelled]
    W2 --> ST
    WN --> ST
    ST --> C[State counts]
```

Table 29.1 – Task Priority Levels

Priority	Use	Notes
Negative / low	Fast, foundational probes	Run first to unblock dependents
0 (default)	Standard module tasks	FIFO within the same priority
High positive	Deferred / enrichment work	Run last
Sentinel (`2^31-1`)	Worker shutdown marker	Never counted as a task

Each `Task` records `created_at`, `started_at`, `finished_at`, and derives `duration`. Unstarted coroutines are explicitly closed (`Task.close()`) to prevent "coroutine was never awaited" warnings and resource leaks.

---

30. Worker Pool and Concurrency

The `WorkerPool` consumes tasks from the scheduler with a fixed number of asyncio workers (default 8, configurable via `scan.max_workers`). It applies a per-task timeout, isolates failures, and supports cancellation of both queued and in-flight work.

Figure 30.1 – Worker Pool Concurrency Model

```
flowchart TD
    Start[pool.run] --> S[send N sentinels]
    S --> Spawn[spawn N workers]
    Spawn --> Loop{next_task}
    Loop -->|Task| Exec[execute with timeout]
    Loop -->|sentinel| Exit[worker exits]
    Exec --> OK[mark_done]
    Exec --> TO[mark_failed: timeout]
    Exec --> ERR[mark_failed: error]
    Exec --> CAN[mark_cancelled]
    OK --> Loop
    TO --> Loop
    ERR --> Loop
    CAN --> Loop
    Exit --> Gather[gather all workers]
    Gather --> Stats[PoolStats]
```

The pool exposes `peak_in_flight` for observability and a `PoolStats` summary (`submitted`, `done`, `failed`, `cancelled`, `duration`). The `on_task_done` callback — used by the CLI and GUI for live progress — is wrapped so that a misbehaving callback can never crash a worker.

---

31. Caching Subsystem

`AsyncTTLCache` is a single-event-loop LRU cache with per-key TTL. It supports an enable flag, a default TTL, a maximum size with LRU eviction, and hit/miss/eviction/expiration counters.

Table 31.1 – Cache Configuration Defaults

Setting	Default	Meaning
`cache.enabled`	`true`	Master switch
`cache.ttl`	`300.0` s	Default time-to-live
`cache.max_size`	`1024`	Maximum entries before LRU eviction

Modules memoise expensive lookups through `ctx.cached(key, factory, ttl=…)`, which returns a cached value or runs the factory and stores the result. The HTTP client additionally uses the cache for conditional requests (`If-None-Match` / `If-Modified-Since`), returning a stored payload on `304 Not Modified`. The benchmark suite records cache throughput above 2.5 million operations per second in memory (Chapter 67).

---

32. Offload Pool and Blocking Operations

Blocking work — PDF/EXIF parsing, large JSON decode, filesystem reads — must not run on the event loop, or it stalls every concurrent task. `OffloadPool` wraps a bounded `ThreadPoolExecutor` (default 4 workers, `performance.offload_workers`) so callers can `await pool.run(func, …)` without spawning unbounded threads.

Modules use `ctx.to_thread(...)`, which prefers the per-scan pool and falls back to `asyncio.to_thread`. A process-wide convenience pool (`offload()`, `get_offload_pool()`) serves scripts and tests. The pool tracks `submitted` / `completed` counters and shuts down cleanly via `shutdown()` or `aclose()`.

---

33. Configuration Management

Configuration is layered with clear precedence and dotted-key access.

Table 33.1 – Configuration Precedence

Priority	Source	Example
1 (lowest)	Built-in defaults (`DEFAULTS`)	`scan.max_workers = 8`
2	JSON file (`config/settings.json`)	`{"scan": {"max_workers": 16}}`
3 (highest)	`.env` file and environment variables	`INTELXTRACT_SCAN__MAX_WORKERS=16`

Environment variables use the `INTELXTRACT_` prefix; a double underscore maps to nesting (`SCAN__MAX_WORKERS` → `scan.max_workers`). Values are coerced to JSON types when possible. The `api_keys` section is **never** written back to the config file (`Config.save()` strips it); secrets belong to the encrypted vault (Chapter 60). `load_env_file()` reads `KEY=VALUE` pairs, honours quotes and comments, and never overwrites variables that are already set in the environment.

---

34. Logging and Audit Subsystem

Logging uses the standard library with a dedicated `intelxtract` namespace. Handlers are a console stream (optional) and a rotating file handler (`logging.max_bytes` , `logging.backup_count`).

Two features distinguish the logging layer:

1. **Secret redaction.** A custom formatter applies regular expressions that mask bearer tokens and `key=value` / `key: value` pairs for `api_key` , `token` , `secret` , `password` , and `authorization` , replacing the value with `***REDACTED***` .
2. **Scan context.** A `ContextVar` holds the current scan id; a logging filter injects it into every record as `[scan=<id>]` , so log lines from concurrent modules remain attributable. The `scan_context()` context manager binds and restores the id around each scan.

The database additionally stores structured audit events in the `logs` table (level, message, source, JSON context, timestamp), queryable via the `LogRepository` .

---

35. Database Design

IntelXtract persists all state in a single SQLite database accessed asynchronously through `aiosqlite`. The schema is normalised around **targets** → **scans** → **findings**, with satellite tables for reports, schedules, cases, plugins, keys, settings, logs, and history.

Table 35.1 – Database Tables

Table	Purpose	Key columns
`schema_version`	Applied migrations	`version` (PK), `name`, `applied_at`
`targets`	Classified indicators	`id`, `value`, `type`; unique `(value, type)`
`scans`	Scan runs	`id`, `target_id` FK, `uuid` (unique), `mode`, `status`, `runs_json`
`findings`	Normalised intelligence	`id`, `scan_id` FK, `module`, `title`, `severity`, `confidence`, `data`, `evidence`, `content_hash`
`reports`	Generated report files	`id`, `scan_id` FK, `path`, `format`
`plugins`	Plugin metadata	`name` (unique), `version`, `api_version`, `enabled`, `source`, `error`
`api_keys`	Encrypted secrets	`provider` (unique), `ciphertext`, `hint`
`logs`	Audit events	`level`, `message`, `source`, `context`
`history`	Per-target change events	`target_id`, `scan_id`, `event_type`, `payload`
`settings`	Runtime settings	`key` (PK), `value` (JSON), `updated_at`
`schedules`	Recurring scans	`target`, `mode`, `interval_seconds` / `cron`, `enabled`, `status`, `next_run_at`
`cases`	Investigations	`name` (unique), `description`, `status`
`case_notes`	Case notes	`case_id` FK, `body`
`case_evidence`	Case evidence links	`case_id`, `scan_id`, `finding_id`, `kind`, `label`

Referential integrity is enforced with `ON DELETE CASCADE` for scan-owned rows (findings, reports) and `ON DELETE SET NULL` where a soft link is preferable (for example, `history.scan_id`, `case_evidence.scan_id`). Check constraints guard `confidence ∈ [0,1]` and boolean-like `enabled` columns. Indices cover every foreign key and the common filters (`scans.status`, `findings.scan_id`, `schedules.next_run_at`, `cases.name`).

35.1 Entity-Relationship Diagram

Figure 35.1 – Entity-Relationship Diagram

```
erDiagram
    TARGETS ||--o{ SCANS : "has"
    SCANS ||--o{ FINDINGS : "produces"
    SCANS ||--o{ REPORTS : "generates"
    TARGETS ||--o{ HISTORY : "records"
    SCANS ||--o{ HISTORY : "referenced by"
    SCANS ||--o{ SCHEDULES : "last scan"
    CASES ||--o{ CASE_NOTES : "has"
    CASES ||--o{ CASE_EVIDENCE : "attaches"
    SCANS ||--o{ CASE_EVIDENCE : "linked"
    FINDINGS ||--o{ CASE_EVIDENCE : "linked"
    PLUGINS {
        text name
        int api_version
        bool enabled
    }
    API_KEYS {
        text provider
        text ciphertext
        text hint
    }
    SETTINGS {
        text key
        text value
    }
    LOGS {
        text level
        text message
    }
```

---

36. Database Migrations

The schema evolves through **forward-only, immutable migrations**. Each migration has a unique version and name and is recorded in `schema_version` once applied. Migration 1 loads `database/schema.sql`; later migrations add columns or tables via `ALTER TABLE` / `CREATE TABLE`.

Table 36.1 – Migration History

Version	Name	Change
1	initial schema	targets, scans, findings, reports, plugins, api_keys, logs, history, settings
2	finding normalized fields	add `findings.title`, `evidence`, `content_hash`, `collected_at`
3	recurring schedules	add `schedules` table + indices
4	case management	add `cases`, `case_notes`, `case_evidence`

The runner validates that versions are unique and contiguous starting at 1, applies only pending migrations in order, and records the failure point without marking a failed migration as applied. The initial SQL is loaded with `importlib.resources` so it resolves correctly inside a PyInstaller bundle.

---

37. Repository and Persistence Layer

All SQL lives in `database/repositories.py` (the only place besides migrations). Rows are returned as **frozen, slotted dataclasses**; JSON columns are parsed on read and serialised on write. This keeps SQL out of the engine and the UI and gives the rest of the application typed records.

The `Repositories` bundle exposes:

- `TargetRepository` — `get_or_create`, `get`, `list`
- `ScanRepository` — `create`, `create_for_result`, `list`, `list_detailed`, `latest_for_target`, `stats`
- `FindingRepository` — `bulk_create`, `list_for_scan`, `count_for_scan`, `get`
- `ReportRepository`, `PluginRepository`, `ApiKeyRepository`, `LogRepository`, `HistoryRepository`
- `ScheduleRepository` — implements the `ScheduleStore` protocol consumed by the runner
- `CaseRepository` — cases plus notes and evidence
- `SettingRepository` — JSON settings

`database/persistence.py` bridges the engine and the database **without the engine importing the database**: `make_result_sink(db)` returns an async callable that persists a completed `ScanResult` (target, scan, findings). It also implements change history (`record_scan_change`) by diffing consecutive scans, and assembles case payloads (`collect_case_payload`) for combined reporting, de-duplicating findings by id.

---

38. Module Architecture

Every collector subclasses `BaseModule` (in `modules/base.py`) and declares three pieces of metadata: `name`, `target_types`, and `requires_keys`. The engine's planner skips modules whose `requires_keys` are unconfigured, and modules whose `target_types` do not include the classified type.

```
class BaseModule(ABC):
    name: str = ""
    target_types: tuple[TargetType, ...] = ()
    requires_keys: tuple[str, ...] = ()
    description: str = ""

    def validate(self, target: str) -> bool: ...

    @abstractmethod
    async def run(self, target: str, ctx: ModuleContext) -> ModuleResult: ...
```

Shared conveniences (`http_get_json`, `http_get_text`, `http_fetch`) keep collectors thin, and a `retry` decorator plus `run_with_timeout` provide resilience. Collectors return a `ModuleResult` whose `findings` are normalised by `parse_findings` and deduplicated by a stable SHA-256 `content_hash` over module, title, severity, data, and evidence.

Table 38.1 – OSINT Module Inventory

#	Module	Target type	Requires key	Description
1	`dns`	Domain	—	DoH records A/AAAA/CNAME/NS/MX/TXT, SPF, DMARC, DKIM
2	`http`	Domain	—	HTTP(S) status, redirects, robots/sitemap, security headers
3	`ssl`	Domain	—	TLS certificate metadata (subject, issuer, validity, SANs)
4	`whois`	Domain	—	RDAP registrar, dates, contacts, DNSSEC
5	`subdomain`	Domain	—	Passive subdomains via certificate transparency
6	`geo`	IP	—	Geolocation and network facts (cached)
7	`rdns`	IP	—	Reverse DNS (PTR) + forward-confirmed lookup
8	`reputation`	IP	AbuseIPDB (opt.)	DNSBL (Spamhaus ZEN) + optional AbuseIPDB
9	`favicon`	URL	—	Favicon fingerprint (FNV-1a over base64 md5)
10	`headers`	URL	—	Response headers, cookie flags, disclosures
11	`http_methods`	URL	—	Allowed methods, page title, content type
12	`robots`	URL	—	robots.txt groups and sitemap extraction
13	`tech`	URL	—	Tech-stack fingerprinting
14	`email`	Email	—	Validation, mail-domain MX/SPF/DMARC, Gravatar
15	`breach`	Email	HIBP	Breach exposure (key-gated)
16	`username`	Username	GitHub (opt.)	Presence across 30+ sites + enrichment
17	`certificate`	Domain	—	Certificate chain, TLS probe, CT history
18	`metadata`	File	—	PDF / image EXIF / OOXML office properties
19	`news`	Domain	—	News timeline from Google News RSS

39. Domain Reconnaissance Modules

Table 39.1 – Domain Module Findings

Module	Representative findings	Notable fields
`dns`	`DNS: A|AAAA|CNAME|NS|MX|TXT records`, `DNS: SPF record`, `DNS: no SPF record`, `DNS: DMARC record`, `DNS: no DMARC record`, `DNS: DKIM selector '<sel>'`	`type`, `records`, `record`, `all`, `policy`
`http`	`HTTP: https responds`, `HTTP: redirect chain`, `HTTP: https unreachable`, `HTTP: <label> present/unreachable`, `HTTP: missing <header>`	`status`, `chain`, `robots`, `sitemap`
`ssl`	`SSL: certificate`, `SSL: subject alternative names`, `SSL: self-signed certificate`, `SSL: certificate expired`	`subject_cn`, `issuer`, `subject_alt_names`, `not_after`
`whois`	`RDAP: registrar`, `RDAP: registration dates`, `RDAP: nameservers`, `RDAP: status flags`, `RDAP: DNSSEC signed/not signed`, `RDAP: <role> contact`	`registrar`, `nameservers`, `email`, `org`
`subdomain`	`Subdomains: certificate transparency`	`subdomains`
`news`	`News: timeline`, `News: no recent articles found`	`articles[]`
`certificate`	`Certificate: chain details`, `Certificate: subject alternative names`, `Certificate: transparency history`	`serial`, `issuer`, `subject`, `subject_alt_names`, `related_names`

The `dns` module queries DNS-over-HTTPS and probes a configurable DKIM selector list. The `ssl` module opens a TLS connection and parses the leaf certificate, flagging expiry and self-signing — the two behaviours that the risk engine weighs most heavily. The `subdomain` and `certificate` modules both consume certificate-transparency data (`crt.sh`), giving passive discovery without active probes.

---

40. IP Intelligence Modules

Table 40.1 – IP Module Findings

Module	Representative findings	Notes
`geo`	`Geo: IP location`, `Geo: network`, `Geo: lookup unavailable`	ip-api.org, cached 24 h
`rdns`	`RDNS: PTR record`, `RDNS: no PTR record`, `RDNS: forward-confirmed`, `RDNS: forward-confirmation failed`	DoH PTR + forward confirmation (spoof resistance)
`reputation`	`Reputation: DNSBL listing`, `Reputation: DNSBL clean`, `Reputation: AbuseIPDB abusive/clean`, `Reputation: IPv6 out of scope for DNSBL`	Spamhaus ZEN; AbuseIPDB optional

Forward-confirmation is an important correctness measure: a PTR record alone is untrustworthy, so the module resolves the PTR hostname back to an address and only reports a match when it round-trips.

---

41. URL and Website Modules

Table 41.1 – URL and Website Module Findings

Module	Representative findings
`headers`	`Headers: <name> present/missing`, `Headers: server disclosure`, `Headers: insecure cookie`, `Headers: response compression`, `Headers: response profile`
`http_methods`	`HTTP: allowed methods`, `HTTP: TRACE enabled`, `HTTP: OPTIONS restricted`, `HTTP: page title`, `HTTP: page profile`
`favicon`	`Favicon: fingerprint`, `Favicon: not found`
`robots`	`Robots: parsed robots.txt`, `Robots: no robots.txt`, `Robots: sitemap URLs`
`tech`	`Tech: <name>` per detected technology

The headers module reports missing security headers (`strict-transport-security`, `content-security-policy`, `x-frame-options`, `x-content-type-options`, `referrer-policy`, `permissions-policy`) and insecure cookie flags — each of which maps to a documented risk rule. The favicon fingerprint (FNV-1a over the base64 MD5 of the icon) enables infrastructure clustering, a classic OSINT technique.

---

42. Email Intelligence Modules

Table 42.1 – Email Module Findings

Module	Representative findings
`email`	`Email: disposable address`, `Email: malformed address`, `Email: MX records`, `Email: domain accepts no mail (no MX)`, `Email: SPF policy`, `Email: no SPF record`, `Email: DMARC policy`, `Email: no DMARC record`, `Email: Gravatar profile`
`breach`	`Breach: address exposed in known breach(es)`, `Breach: no known breaches`, `Breach: API key rejected`, `Breach: check unavailable`

The `email` module performs syntactic validation, queries the mail domain's MX/SPF/DMARC records, checks the address against a disposable-provider list, and probes Gravatar. The `breach` module is key-gated behind Have I Been Pwned; without a configured key it is skipped by the planner rather than failing.

---

43. Username Reconnaissance Module

The `username` module checks a handle against 30+ public sites using a declarative pattern table and a paced concurrent probe (`_PacedProbe`) that respects a per-site interval to avoid overwhelming targets. It reports `Username: exists on <site>` per positive hit, a `Username: profile summary` , and optional GitHub enrichment (`Username: GitHub profile enrichment`) that contributes a `social_profile` entity to correlation. A `Username: all site verdicts unknown` finding is emitted when the network is unavailable, avoiding false negatives.

---

44. Certificate Transparency Module

The `certificate` module collects the full chain, parses subject/issuer/SANs, and queries certificate-transparency history, emitting `Certificate: chain details`, `Certificate: subject alternative names`, and `Certificate: transparency history`. It also exposes a TLS probe with protocol/cipher reporting. Certificate data is a rich source of sibling-domain and organisation entities, which is why it participates in correlation with a high edge confidence.

---

45. File Metadata Modules

The `metadata` module parses document properties from three families:

- **PDF** (via `pypdf`) — author, producer, creation/modification dates, XMP.
- **Images** (via `Pillow`) — EXIF tags including GPS coordinates.
- **Office OOXML** (docx/xlsx/pptx) — core and application properties.

It emits a per-family finding (`Metadata: unsupported file type`, `Metadata: unparseable document`) and, critically, `Metadata: GPS coordinates` when EXIF carries location — the highest-weighted single exposure rule (10 points). Parsing runs through the offload pool because it is CPU/IO-bound.

---

46. News and Timeline Module

The `news` module builds a recent-news timeline from Google News RSS and any configured feeds per domain. It emits `News: timeline` (with an `articles[]` array of title, URL, publisher, and publish date) or `News: no recent articles found`. The report builder consumes these articles to construct a chronological timeline (Chapter 52), and correlation treats publishing hosts as `mentions` edges.

---

47. Correlation Engine

The correlation engine turns a flat list of findings into a connected entity graph using NetworkX's `MultiDiGraph`. Node identity comes from `core/entities.py` (type + normalised value, hashed to `kind:<12-hex>`); edges carry a semantic type and a confidence.

Figure 47.1 – Correlation Graph Structure

```
flowchart LR
    T((target: domain)) -->|resolves_to 0.9| IP1((ip))
    T -->|registered_by 0.9| ORG((organization: registrar))
    T -->|issued_to 0.7| ISSUER((organization: CA))
    T -->|part_of 0.9| SUB((subdomain))
    CERT((certificate)) -->|issued_to 0.8| ISSUER
    T -->|issued_to 0.9| CERT
    MAIL((email)) -->|same_email 0.9| T
    T -->|uses 0.8| TECH((technology))
    U((username)) -->|profile 0.9| PROF((social_profile))
    PUB((publisher domain)) -->|mentions 0.4| T
```

Table 47.1 – Correlation Edge Types

Constant	Edge type	Typical confidence	Example
`EDGE_RESOLVES_TO`	resolves_to	0.4–0.9	domain → IP (A/AAAA), domain → CNAME alias
`EDGE_REGISTERED_BY`	registered_by	0.9	domain → registrar organisation
`EDGE_ISSUED_TO`	issued_to	0.7–0.9	domain → issuer / certificate → issuer
`EDGE_SAME_EMAIL`	same_email	0.7–0.9	email → domain, email → username
`EDGE_PART_OF`	part_of	0.9	base domain → subdomain
`EDGE_MENTIONS`	mentions	0.4	publisher domain → target
`EDGE_PROFILE`	profile	0.9	username → social profile
`EDGE_USES`	uses	0.8	domain → technology

Figure 47.2 – Correlation Edge Types

```
flowchart TD
    E1[resolves_to] --> M{DNS applicator}
    E2[registered_by] --> W{WHOIS applicator}
    E3[issued_to] --> S{SSL/certificate applicator}
    E4[same_email] --> EM{Email linkage}
    E5[part_of] --> SD{Subdomain applicator}
    E6[mentions] --> N{News applicator}
    E7[profile] --> U{Username applicator}
    E8[uses] --> T{Tech applicator}
```

When the same edge is added twice, the engine keeps the **maximum** confidence. The graph is fully deterministic — nodes and edges are sorted by stable keys — so exports, persistence, and diffs are byte-stable. If a scan produces no entities, `build_graph` returns `None` rather than an empty graph.

---

48. Entity Model

Entities are the lowest common denominator across modules: an IP surfaced by both a DNS A record and a certificate entry is one node. Each module declares extractors against `Finding.data`; the `EntityBag` aggregates occurrences into a single entity with a count and a set of contributing sources.

Table 48.1 – Entity Kinds

Kind	Normalisation
`domain`	lowercase, strip trailing dot / `*.`
`subdomain`	lowercase, strip leading `*.`
`email`	lowercase
`username`	strip leading `@`, lowercase
`ip`	canonical `ipaddress` form
`organization`	whitespace-collapsed
`certificate`	whitespace-collapsed, uppercase (serial/identity)
`document`	lowercase filename
`social_profile`	URL verbatim
`technology`	lowercase

Entity ids are `kind:<first-12-hex-of-sha256(normalised value)>`, giving stable, compact identifiers suitable for graph keys and JSON export. `emit()` validates candidates (for example, rejecting domains without a dot, or emails without `@`) so junk never enters the graph.

---

49. Risk Engine

The risk engine (`core/risk.py`) is a **pure function**: identical findings always yield identical output. It walks every finding, applies the rule table, de-duplicates identical (rule, finding) pairs, sorts by contribution, and computes per-category sub-scores and a capped total.

```
score = min(100.0, sum(min(category_score, ceiling[category]) for category in categories))
```

Each matched rule is recorded with its `rule_id` , `category` , `label` , `points` , and the finding it fired on, so the score is fully traceable. A human-readable `summary` lists the overall rating and the top contributing rules.

---

50. Risk Rules Catalog

Table 50.1 – Risk Rule Catalog

Rule id	Category	Weight	Module	Trigger
`tls_cert_expired`	TLS	35	ssl	Certificate expired
`tls_self_signed`	TLS	18	ssl	Self-signed certificate
`tls_no_https`	TLS	10	http	HTTPS unreachable
`headers_trace`	HEADERS	15	http_methods	TRACE enabled
`headers_hsts`	HEADERS	9	headers	HSTS missing
`headers_csp`	HEADERS	9	headers	CSP missing
`headers_cookie_no_secure`	HEADERS	7	headers	Cookie without Secure
`headers_xframe`	HEADERS	4	headers	X-Frame-Options missing
`headers_server`	HEADERS	4	headers	Server disclosed
`headers_xcto`	HEADERS	3	headers	X-Content-Type-Options missing
`headers_referrer`	HEADERS	2	headers	Referrer-Policy missing
`headers_cookie_weak`	HEADERS	2	headers	Cookie missing HttpOnly/SameSite
`headers_permissions`	HEADERS	2	headers	Permissions-Policy missing
`dns_no_dmarc`	DNS	6	dns	No DMARC record
`dns_no_spf`	DNS	4	dns	No SPF record
`dns_dnssec_absent`	DNS	3	whois	DNSSEC not signed
`dns_no_mx`	DNS	5	email	No MX records
`dns_spf_allow_all`	DNS	6	email	SPF `+all`
`dns_dmarc_none`	DNS	6	email	DMARC `p=none`
`exposure_breach`	EXPOSURE	20	breach	Address in breach
`exposure_gps`	EXPOSURE	10	metadata	GPS in document metadata
`exposure_disposable`	EXPOSURE	4	email	Disposable provider

Table 50.2 – Risk Category Ceilings

Category	Rules	Ceiling
TLS	3	40
HEADERS	10	30
DNS	6	20
EXPOSURE	3	25

A few rules are gated by predicates on the finding's data (for example, `dns_spf_allow_all` only fires when the SPF policy is exactly `+all`, and the two cookie rules are mutually exclusive based on whether the `Secure` flag is present). This makes the catalog precise rather than string-matching the web.

---

51. Risk Rating and Verdicts

Table 51.1 – Risk Rating Bands

Rating	Score threshold	Verdict
Critical	≥ 75	Immediate remediation required
High	≥ 50	Significant weaknesses found
Medium	≥ 25	Moderate weaknesses found
Low	≥ 0	Low overall exposure

The bands are evaluated highest-first, so the first matching threshold wins. A scan with no contributing findings returns a score of 0 and a LOW rating with an explicit summary line.

52. Reporting Subsystem

All exporters consume a single shared model, `ReportModel`, built by `reports/builder.py`. The builder re-derives the correlation graph and risk score from the findings when they are not supplied, so reports render correctly even for scans persisted before those phases existed. `ReportModel` contains: `generated_at`, `app`, `scan`, `target`, `findings`, `correlation`, `risk`, `severity_counts`, `module_counts`, `timeline`, `executive_summary`, `evidence_refs`, and `methodology`.

Because every format renders from the same model, JSON, CSV, Markdown, HTML, and PDF outputs remain mutually consistent. The executive summary selects the top three findings by severity; the evidence references cap at six module/title/evidence triples; and the timeline interleaves scan start/finish events with news article publication dates.

---

53. Report Templates and Exporters

Five exporters share the `ExportResult(format, data, warning)` contract. Formats are selected case-insensitively; `md` / `markdown` / `mdown` are aliases.

Table 53.1 – Report Formats

Format	Producer	Structure	Caveats
JSON	`json.dumps`	Full canonical model, 2-space indent, `default=str`	Machine-consumable; no lossy projection
CSV	`csv.writer`	Row per finding: severity, module, title, confidence, evidence	Evidence is truncated for tabular readability
Markdown	string builder	Headings, tables, timeline, methodology	Escapes `|` and newlines in cells
HTML	Jinja2 (`report.html.j2`)	Branded, self-contained, auto-escaped	Print mode adds page-break styling
PDF	WeasyPrint	Rendered from the HTML template	Falls back to print HTML if WeasyPrint/ system libs absent

Figure 53.1 – Export Pipeline

```
flowchart LR
    M[ReportModel] --> J[JSON exporter]
    M --> C[CSV exporter]
    M --> MD[Markdown exporter]
    M --> H[HTML exporter]
    H --> P[WeasyPrint]
    P -->|available| PDF[PDF bytes]
    P -->|missing libs| FB[print HTML + warning]
    BRAND[brand_tokens] -.-> H
    TPL[report.html.j2] -.-> H
```

The HTML template consumes `brand_tokens()` derived from `BRAND.md`, keeping colours/identity centralised. HTML auto-escaping is enabled for `html`, `j2`, and `xml` so hostile finding content cannot inject script (threat T2). PDF export imports WeasyPrint lazily via `importlib`; if either the package or its system libraries (Pango, Cairo) are missing, it transparently writes a print-friendly HTML and returns an explicit warning rather than failing — a deliberate graceful degradation documented for the user.

---

54. Command-Line Interface

The CLI is built on Typer with Rich rendering. The root command is `intelxtract`; six sub-applications group related commands. It provides full functional parity with the GUI for headless and automation use.

Table 54.1 – CLI Command Reference

Command	Purpose	Key options
`scan <target>`	Run a scan and persist it	`--mode/-m`, `--modules/-M`, `--format/-f console|json`, `--db`
`report <scan_id>`	Render a saved scan	`--format json|html|pdf|csv|md`, `--out`, `--db`
`explain <scan_id>`	AI/rule-based narrative	`--kind summary|next-steps|finding|compare`, `--finding`, `--compare`, `--json`
`history list`	List scans	`--target/-t`, `--status/-s`, `--db`
`history show <scan_id>`	Scan detail	`--db`
`config get/set/list/path`	Inspect/modify config	dotted keys (`config get cache.ttl`)
`key list/set/get/delete`	Manage encrypted keys	`--value`, `--reveal`, `--yes/-y`
`plugin list/info/enable/disable`	Manage plugins	`--db`
`schedule add/list/remove/enable/disable/run`	Recurring scans	`--every`, `--cron`, `--mode`, `--disabled`, `--once`, `--iterations`, `--poll`
`case create/list/show/note/attach/export`	Investigations	`--description`, `--status`, `--db`
`doctor`	Environment health check	—
`update`	Check PyPI for a newer version	—
`--version`	Print version	—

The root callback configures logging (with `-v` switching to DEBUG) and supports `--version`. `doctor` runs a sequence of environment checks (Python version, virtual-environment presence, config readability, dependency availability) and prints a pass/fail report. `update` compares the running version against the latest on PyPI without sending identifying information.

---

55. GUI Architecture

The desktop client is built with PySide6 (Qt 6) and drives the **same engine** as the CLI on a background thread, keeping the interface responsive during scans.

Figure 55.1 – GUI Architecture

```
flowchart TB
    subgraph UI["Main Window (Qt main thread)"]
        NAV[Navigation / Sidebar]
        PAGES[13 Pages]
        VIZ[Visualisations]
    end
    subgraph Bridge["EngineBridge (QThread)"]
        SIG[Qt signals: started / progress / finished / moduleDone / log]
    end
    ENGINE[ScanEngine + WorkerPool]
    DB[(SQLite)]
    CTX[AppContext]
    UI <-->|slots & signals| Bridge
    Bridge --> ENGINE
    ENGINE --> DB
    CTX -.-> UI
    CTX -.-> ENGINE
    DB_SYNC[db_sync] --> UI
    CASE_SYNC[case_sync] --> UI
```

`EngineBridge` owns a `QThread` that runs the asyncio event loop; the engine's synchronous callbacks (`on_module_done`, progress) are marshalled to the UI thread via Qt signals, so no widget is ever touched from a worker thread. `gui/context.py` (`AppContext`) wires configuration, database, and registries; `gui/db_sync.py` and `gui/case_sync.py` keep list views in sync with persistence; `gui/theme.py` defines the visual theme.

---

56. GUI Pages

Table 56.1 – GUI Pages

Page	Function
Dashboard	Overview: recent scans, risk summary, quick actions
Quick Scan	One-click profile scan (quick mode)
Deep Scan	Full-profile scan with module selection
Results	Findings table with severity filters and detail pane
History	Browse and reopen past scans
Reports	Generate and export reports in any format
Cases	Create investigations, add notes, attach evidence, export combined case report
Schedules	Manage recurring scans; run scheduler
Plugins	Enable/disable plugins, view manifest metadata
API Manager	Manage encrypted API keys
Assistant	Opt-in AI narrative/explanations
Settings	Edit configuration values
About	Version, licence, and project links

`scan_page.py` and `base.py` provide shared scan-page scaffolding consumed by the quick and deep scan pages.

Figure 56.1 – GUI Navigation Flow

```
flowchart LR
    D[Dashboard] --> Q[Quick Scan]
    D --> R[Results]
    Q --> R
    R --> REP[Reports]
    D --> H[History]
    D --> C[Cases]
    D --> S[Schedules]
    SET[Settings] --> API[API Manager]
    D --> A[Assistant]
    D --> P[Plugins]
```

The following figures are screenshot placeholders to be populated from a running instance before submission:

- **Figure 56.2 – Dashboard Page** *(screenshot placeholder)*
- **Figure 56.3 – Quick Scan Page** *(screenshot placeholder)*
- **Figure 56.4 – Results Page** *(screenshot placeholder)*
- **Figure 56.5 – Correlation Graph Visualisation** *(screenshot placeholder)*
- **Figure 56.6 – Cases Page** *(screenshot placeholder)*
- **Figure 56.7 – Reports Page** *(screenshot placeholder)*
- **Figure 56.8 – API Manager Page** *(screenshot placeholder)*
- **Figure 56.9 – Scheduler Page** *(screenshot placeholder)*

---

57. GUI Visualizations

Table 57.1 – Visualisation Widgets

Widget	Module	Description
Severity/Module charts	`viz/charts.py`	Bar/donut summaries of findings by severity and module
Correlation graph	`viz/graph_widget.py`	Interactive NetworkX entity graph rendering
Timeline	`viz/timeline.py`	Chronological scan/news events
World map	`viz/world_map.py`	Geographic plotting of IP geolocation findings

Figure 57.1 – Visualization Data Flow

```
flowchart LR
    SCAN[ScanResult] --> CH[charts: severity/module counts]
    SCAN --> GR[graph_widget: entities + edges]
    SCAN --> TL[timeline: scan + news events]
    GEO[geo findings] --> WM[world_map: coordinates]
```

Charts are rendered without external plotting dependencies by painting directly with Qt, keeping the dependency footprint small. The graph widget lays out the correlation MultiDiGraph, the timeline orders events by timestamp, and the world map plots geolocation points on a bundled equirectangular base map.

---

58. AI Assistant

The AI assistant is **disabled by default** and strictly opt-in. It has two layers: a deterministic, offline analysis path that always works, and an optional OpenAI-compatible chat path that activates only when a user configures an endpoint.

Table 58.1 – Assistant Modes

Mode	Provider	Network	Description
Off (default)	`DisabledAssistant`	None	Returns `DISABLED_WARNING`; no data leaves the host
Deterministic	rule-based `Assistant`	None	Summary, next-steps, finding, and comparison narratives
Remote	`OpenAICompatibleAssistant`	HTTPS to user endpoint	Sends the report payload to `/chat/completions`

The assistant exposes `AssistantAnswer` objects labelled `AI-generated`, and `build_default_provider()` selects the provider from configuration. The provider protocol (`AssistantProvider`) returns `AssistantResponse` / `ChatMessage` values. Because the remote path is opt-in, the threat model (T10) treats it as an explicit, disclosed egress. The CLI `explain` command and the Assistant GUI page share this service; when disabled, `explain` still produces the deterministic answer.

---

59. Plugin SDK

Plugins extend IntelXtract with new collectors without touching the core. A plugin lives in its own directory and declares a manifest plus a module class.

Table 59.1 – Plugin Package Layout

File	Purpose
`manifest.json`	Name, version, `api_version`, entry, description
`plugin.py`	`PluginBase` subclass returning `BaseModule` instances

`PluginBase` subclasses declare their modules; `PluginModuleAdapter` wraps each contributed module so the registry treats it identically to a built-in collector. The loader validates the manifest against `MANIFEST_SCHEMA`, enforces `PLUGIN_API_VERSION`, imports the module in isolation, and records failures in the `plugins` table (with an `error` column) without crashing the application. Plugins are explicitly **trusted code** run with the application's privileges — a residual risk accepted and disclosed.

Two example plugins ship in the repository: **Wayback** (historical snapshots from the Internet Archive) and **VirusTotal** (key-gated reputation). A `_template` plugin demonstrates the minimal structure.

Figure 59.1 – Plugin Lifecycle

```
flowchart LR
    DIR[Plugin directory] --> MAN[Read manifest.json]
    MAN --> VAL{Schema + api_version valid?}
    VAL -->|no| DIS[Disabled + error recorded]
    VAL -->|yes| IMP[Import plugin.py]
    IMP --> ADP[PluginModuleAdapter]
    ADP --> REG[ModuleRegistry]
    REG --> SCAN[Available to scans]
```

60. API Key Vault and Secrets Management

API keys are encrypted at rest with **Fernet** (symmetric authenticated encryption from `cryptography`). The vault key itself is resolved from, in order: the `INTELXTRACT_VAULT_KEY` environment variable, the OS keychain via `keyring`, or a `0600` key file.

Figure 60.1 – Vault Key Resolution and Key Precedence

```
flowchart TB
    subgraph KeyOfKeys["Vault key resolution"]
        ENV[INTELXTRACT_VAULT_KEY] --> KR[OS keyring]
        KR --> KF[0600 key file]
    end
    subgraph Precedence["Runtime API key precedence"]
        E[env / .env] --> V[vault]
        V --> C[config]
    end
```

`vault.py` exposes `mask()` (for safe display) and `hydrate_api_keys()` (to inject resolved keys into module contexts at scan time). The runtime precedence is **env/ `.env` > vault > config**, so an operator can override a stored key per environment without decrypting the vault. Keys are redacted from logs by the logger's redaction rules and are excluded from reports by construction. `key get` masks by default; `--reveal` prints the plaintext only on explicit request.

---

61. Case Management

A *case* groups scans and findings into a single investigation (migration 4: `cases`, `case_notes`, `case_evidence`). Cases support a name, description, and status (open/closed), free-text notes, and evidence links to either a whole scan or a specific finding.

Figure 61.1 – Case Entity Relationships

```
flowchart LR
    CASE((case)) --> NOTE[case_note]
    CASE --> EV[case_evidence]
    EV -->|kind=scan| SCAN[scan]
    EV -->|finding link| FIN[finding]
    SCAN --> FIN2[findings]
```

Combined case export aggregates the findings of every attached scan, de-duplicating by finding id (`collect_case_payload`), and renders one report across the investigation — including notes and evidence labels. This turns a series of point-in-time scans into a coherent, exportable dossier. The CLI provides `case create/list/show/note/attach/export`, mirrored by the Cases GUI page.

---

62. Scheduler and Change Detection

The scheduler supports both fixed **intervals** (`--every` seconds) and **cron** expressions (`--cron "minute hour day month weekday"`). A pure `CronExpression` parser and `compute_next_run()` function compute the next fire time deterministically; bounds and field names are validated up front.

Figure 62.1 – Scheduler Loop

```
flowchart TB
    TICK[Scheduler tick] --> DUE{Due schedules}
    DUE -->|none| SLEEP[Sleep poll interval]
    DUE -->|one+| RUN[Run scan]
    RUN --> DIFF[Diff vs previous scan]
    DIFF --> DELTA[FindingDelta: added / removed / changed]
    DELTA --> HIST[Persist history events]
    RUN --> NEXT[Compute next_run_at]
    NEXT --> SLEEP
```

Change detection (`diff_findings`) compares a scan's findings against the previous scan for the same target and produces a `FindingDelta` of added, removed, and changed findings (matched by `content_hash`). This is persisted to the `history` table, giving an audit trail of how a target's posture changed over time. Overlap protection prevents a slow run from being re-triggered before it finishes, and `ScheduleRunner` / `ScheduleStore` keep the loop decoupled from the database.

---

63. Threat Model

The threat model (documented in `docs/threat_model.md`) treats everything beyond the application boundary as untrusted, **including responses from the targets the user chooses to scan**.

Table 63.1 – Threat Summary

#	Threat	Mitigation
T1	Hostile target payload (huge HTML, decompression bomb, malformed cert/EXIF/XML)	Size caps, timeouts, safe parsers, offloaded parsing, no `eval` / `pickle`
T2	XSS / script injection into reports	Jinja2 auto-escape, escaped raw values, CSP meta tag
T3	Secret leakage	Logger redaction, vault as sole key source, reports exclude secrets
T4	Local theft of DB/keys	`0600` / `0700` permissions, Fernet encryption, keyring when available
T5	Supply-chain compromise	Minimal pinned dependencies, stdlib-first, version-checked plugin imports
T6	Runaway resource use	Pool limits, per-host semaphores, bounded retries, scheduler overlap prevention
T7	SSRF via redirects to internal hosts	Block loopback/link-local/metadata ranges by default; explicit opt-in
T8	Rate-limit / ToS violations	Global + per-host limits, robots.txt respect, polite key-gating
T9	Plugin escape	Manifest gating, isolated import failure, explicit enablement; plugins trusted
T10	AI prompt exfiltration	AI disabled by default; opt-in with disclosure of receiving endpoint
T11	Audit gap	Local scan history and logged settings/key changes

Figure 63.1 – Trust Boundaries

```
flowchart LR
    USER[User] --> APP[IntelXtract app]
    APP --> FS[(Local FS: DB, config, logs, exports)]
    APP --> TARGETS[Scan targets: DNS/WHOIS/HTTP/TLS/CT]
    APP --> APIS[Optional APIs: VT, HIBP, AbuseIPDB]
    APP --> AI[Optional AI endpoint]
    classDef untrusted fill:#fdd;
    class TARGETS,APIS,AI untrusted;
```

The design enforces an **ethical and legal boundary**: only publicly available information and authorised APIs; no exploitation, credential attacks, or authentication-bypass code; robots.txt is respected on profile checks; and out-of-scope items (multi-user hardening, traffic anonymisation, anti-forensics) are explicitly listed.

---

64. Privacy and Ethical Considerations

IntelXtract is **local-first**. The privacy statement (`docs/privacy.md`) commits to four principles: local-first storage, no telemetry, no account, and opt-in egress only.

Table 64.1 – Data Egress

Egress	Trigger	Data sent
Scan targets	Every scan	Normal protocol traffic only
Public APIs (CT, geo, RSS, site checks)	Enabled modules	The researched indicator, nothing about the user
Key-gated providers	Key added **and** module enabled	Indicator + that provider's API key
AI endpoint	Explicit opt-in	The report payload, to the user's endpoint
PyPI	`intelxtract update` only	Version metadata

Table 64.2 – Data Stored Locally

Data	Location
Targets & scan history	Local SQLite DB
Findings, correlation, risk	Local SQLite DB
Reports	`exports/`
Settings (no secrets)	`config/settings.json`
Encrypted keys	DB `api_keys` / OS keychain
Logs (redacted)	`logs/`
Cache	`cache/`

The ethical stance acknowledges that scanned data may contain personal data about third parties (public WHOIS contacts, profiles), making the operator responsible for lawful handling. Full removal is achieved by deleting the data directory (`config path`), `config/` , `logs/` , and `cache/` . No identifying telemetry is ever emitted, and all egress can be routed through a user-configured proxy.

---

65. Testing Strategy

Testing follows the project's quality gate: **lint, format, type-check, and test must all pass** before any change is committed.

```
export PATH="$PWD/.venv/bin:$PATH" && black . && ruff check . && mypy . && pytest -p no:cachep
```

Figure 65.1 – Test Pyramid and Gate

```
flowchart TB
    subgraph Gate
        B[black] --> R[ruff]
        R --> M[mypy]
        M --> P[pytest]
    end
    subgraph Pyramid
        E2E[Integration / E2E]
        QT[pytest-qt GUI tests]
        UNIT[Unit tests]
        E2E --> QT --> UNIT
    end
```

The strategy is layered: **unit tests** for pure logic (risk rules, correlation, cron, config precedence, cache), **module tests** with mocked HTTP for every collector, **database tests** covering migrations and repositories, **CLI tests** via Typer's runner, **GUI tests** using `pytest-qt`, and **integration tests** (including an offline end-to-end scan) marked so CI can exclude or include them deliberately. Coverage is enforced with `fail_under=80`, branch coverage, restricted to `core`, `modules`, `reports`, and `plugins`.

---

66. Test Suites and Coverage

Table 66.1 – Test Suite Summary

Suite	Representative files	Focus
Core	`tests/core/test_engine.py`, `test_cache.py`, `test_config.py`, `test_correlation.py`, `test_risk.py`, `test_scheduler.py`, `test_scheduling.py`, `test_vault.py`, `test_offload.py`, `test_worker_pool.py`	Engine, config, correlation, risk, scheduling, vault, concurrency
Modules	`tests/modules/**` (per collector), `test_http_client.py`	Finding correctness, HTTP behaviours, caching, concurrency caps
Database	`tests/database/test_migrations.py`, repository tests	Forward migrations and typed repositories
CLI	`tests/cli/test_cli.py`	Command wiring, output, exit codes
GUI	`tests/gui/test_qtbot.py`, `test_smoke.py`, `test_viz.py`	Widget construction and interactions
Integration	`tests/integration/test_offline_scan.py`	End-to-end offline scan
Assistant / Reports / Plugins	`tests/assistant/**`, `tests/reports/**`, `tests/plugins/**`	Provider selection, golden report, plugin loading
Performance	`tests/core/test_performance_budget.py`, `tests/test_bench.py`	Performance budget + benchmark smoke

Table 66.2 – Measured Results

Metric	Value
Test files	81
Tests passing	756
Branch coverage	89.54%
Coverage gate	`fail_under = 80`

The suite includes golden-file tests for reports (`tests/reports/golden/report.json`) to detect unintended output drift, and a performance-budget test that fails if core operations regress beyond their budget.

67. Performance and Benchmarks

Performance work (Phase 25) targeted four axes: connection pooling, cache integration, retry logic, and offloading, under a strict "no unboundedness" policy.

Table 67.1 – Benchmark Baseline (2026-10-11, Linux, Python 3.14)

Section	Metric	Value
Cache	Ops/second	2,550,909
Cache	Hit rate	1.00
Engine	Scans/second	7,414 (8 modules/scan)
Offload	Ops/second	15,344 (4 workers)

The `HttpClient` owns one keep-alive `aiohttp.TCPConnector` per client with total and per-host connection caps and a DNS cache. The shared `AsyncTTLCache` caches module results and HTTP responses carrying `ETag` / `Last-Modified`, revalidating with `If-None-Match` / `If-Modified-Since` and serving cached bodies on `304`. Retries use exponential backoff with equal jitter, only for idempotent methods (GET/HEAD/OPTIONS/PUT/DELETE) on 408/429/5xx. Blocking PDF/EXIF/OOXML parsing and the username registry load run on a bounded offload pool.

Table 67.2 – Concurrency Configuration

Key	Default	Meaning
`network.pool_limit`	100	Max open connections
`network.pool_limit_per_host`	8	Max connections to one host
`network.max_concurrency`	16	Global in-flight cap (0 = off)
`network.max_per_host`	4	Per-host in-flight cap (0 = off)
`network.rate_limit`	0	Global requests/sec (0 = off)
`network.rate_limit_per_host`	0	Per-provider requests/sec (0 = off)
`network.backoff_base` / `backoff_max`	0.5 / 30	Retry delay growth and cap (s)
`network.http_cache`	true	ETag/conditional response caching
`performance.offload_workers`	4	Bounded per-scan offload pool
`cache.ttl` / `cache.max_size`	300 / 1024	Cache expiry and size

Figure 67.1 – Request Admission Control

```
flowchart LR
    REQ[Request] --> G{global semaphore}
    G --> H{per-host semaphore}
    H --> P{rate limiters}
    P --> SESSION[aiohttp session]
    SESSION -->|408/429/5xx| RETRY[Backoff retry]
    SESSION -->|ETag 304| CACHE[Serve cached body]
```

Ordering is deadlock-free: every request acquires the global slot before the per-host slot before the session call. Semaphores are per-`HttpClient` (per scan), so parallel scans never share or exhaust one another's budgets. Absolute numbers vary by machine and are intended as same-host comparability indicators.

---

68. Packaging and Distribution

IntelXtract is distributed as a Python wheel/sdist, as standalone PyInstaller binaries, and as a Docker image.

Table 68.1 – Distribution Artefacts

Artefact	Builder	Target
Wheel (`.whl`) + sdist (`.tar.gz`)	`python -m build`	`pip install intelxtract`
CLI binary (`intelxtract`)	PyInstaller `intelxtract-cli.spec`	Linux x86_64, no Python needed
GUI binary (`intelxtract-gui`)	PyInstaller `intelxtract-gui.spec`	Linux x86_64
Docker image	`Dockerfile` / `docker-compose.yml`	Reproducible headless runtime
Checksums	`sha256sum` / `sha512sum`	Integrity verification

Build scripts (`scripts/build.sh` , `scripts/build.ps1`) automate the local workflow. The package exposes two console entry points: `intelxtract = "cli.main:app"` and `intelxtract-gui = "gui:main"` . Metadata declares GPL-3.0-or-later licensing, `requires-python >= 3.13` , and a minimal, pinned runtime dependency set (aiohttp, aiosqlite, typer, rich, jinja2, PySide6, cryptography, networkx, Pillow, pypdf), with optional `dev` , `build` , and PDF/keyring extras.

Figure 68.1 – Packaging Pipeline

```
flowchart LR
    SRC[Source tree] --> WHEEL[wheel + sdist]
    SRC --> PYI[PyInstaller specs]
    PYI --> CLI[intelxtract binary]
    PYI --> GUI[intelxtract-gui binary]
    SRC --> DOCKER[Docker image]
    WHEEL --> REL[GitHub Release]
    CLI --> REL
    GUI --> REL
    REL --> SUM[sha256/sha512 checksums]
```

The `# [tool.pyinstaller]` section records the two spec files. System dependencies are explicit: PDF export needs WeasyPrint plus Pango/Cairo/GDK-Pixbuf/fontconfig; the GUI needs Qt6 runtime libraries. Docker runs Linux-native and is usable on macOS/Windows via Docker Desktop.

---

69. CI/CD Pipeline

Continuous integration and release are driven by two GitHub Actions workflows.

Table 69.1 – CI Workflow

Aspect	Configuration
Trigger	push/PR to `main`
Runner	`ubuntu-latest`
Matrix	Python 3.13 and 3.14
Steps	checkout → `uv` setup → venv + deps → Qt libs → ruff → black `--check` → mypy → pytest (`-m "not integration"`) with coverage

Table 69.2 – Release Workflow

Aspect	Configuration
Trigger	tag `v*`
Permissions	`contents: write` , `packages: write`
Timeout	30 minutes
Steps	quality gate → wheel/sdist → PyInstaller CLI binary → GUI binary → binary smoke tests → checksums → draft GitHub Release

Figure 69.1 – CI/CD Flow

```
flowchart LR
    PUSH[push / PR] --> CI[CI quality job]
    CI -->|ruff| R1[ruff]
    CI -->|format| R2[black --check]
    CI -->|types| R3[mypy]
    CI -->|tests| R4[pytest + coverage]
    TAG[tag v*] --> REL[Release job]
    REL --> QG[quality gate]
    QG --> B1[wheel + sdist]
    REL --> B2[PyInstaller CLI]
    REL --> B3[PyInstaller GUI]
    B2 --> SMOKE[binary smoke tests]
    B1 --> GH[draft GitHub Release]
    B3 --> GH
    SMOKE --> GH
    GH --> CS[checksums]
```

The CI job installs the exact Qt system libraries needed for offscreen GUI tests. The release job additionally installs WeasyPrint's system libraries so PDF generation is exercised, builds both binaries, then **smoke-tests the CLI binary** (`--version` , `doctor` , and a real `scan example.com --mode quick --format json`) before publishing a draft release with generated notes and checksums. Versions are pinned with `==` throughout, satisfying the supply-chain goal (T5).

---

70. Comparison with Existing Tools

IntelXtract is positioned between single-purpose reconnaissance tools (which do one thing well) and heavyweight commercial platforms (which require accounts and cloud processing).

Table 70.1 – Feature Comparison

Capability	IntelXtract	Typical `whois`/`dig`	`theHarvester`	Maltego	Shodan
Dual CLI + desktop GUI	Yes	No	No	Desktop GUI	Web
Local-first / no account	Yes	Yes	Yes	Partial	No
Correlation graph	Yes (NetworkX)	No	No	Yes	No
Deterministic risk score	Yes (22 rules)	No	No	Partial	No
Multi-format reports	JSON/CSV/MD/HTML/PDF	No	Text/HTML	Yes	Yes
Email/breach checks	Yes (key-gated)	No	Partial	Via transforms	No
Plugin SDK	Yes	No	No	Yes (transforms)	No
Encrypted key vault	Yes (Fernet)	N/A	No	Yes	N/A
Case management	Yes	No	No	Yes	No
Recurring scans + diff	Yes	No	No	Partial	Partial
AI narrative (opt-in)	Yes	No	No	No	No
Open source	Yes (GPL-3.0+)	Yes	Yes	No	No

Figure 70.1 – Positioning

```
quadrantChart
    x-axis "Low automation" --> "High automation"
    y-axis "Narrow scope" --> "Broad scope"
    quadrant-1 "Commercial platforms"
    quadrant-2 "IntelXtract"
    quadrant-3 "Single-purpose CLI tools"
    quadrant-4 "Scripted pipelines"
    "dig/whois": [0.15, 0.1]
    "theHarvester": [0.35, 0.3]
    "Maltego": [0.75, 0.85]
    "Shodan": [0.8, 0.5]
    "IntelXtract": [0.72, 0.72]
```

IntelXtract's distinguishing combination is **local-first operation with no account or telemetry**, a **deterministic, auditable risk model**, integrated **correlation and case management**, and a **dual CLI/GUI** — without the cloud dependency and per-seat cost of commercial platforms.

---

71. Advantages

- **Local-first and private** — no account, no telemetry, opt-in egress only; all data resides in a user-controlled SQLite database and export directory.
- **Deterministic and auditable** — the risk engine is a pure function; identical findings yield identical scores, and every matched rule is traceable to its finding.
- **Modular and extensible** — 19 built-in collectors behind a uniform `BaseModule` interface plus a manifest-gated Plugin SDK.
- **Full parity interfaces** — CLI and GUI drive the same engine and database, so automation and interactive use are interchangeable.
- **Integrated intelligence pipeline** — collection → entity extraction → correlation → risk scoring → multi-format reporting, plus case management and historical diffing.
- **Resilient networking** — connection pooling, ETag caching, bounded concurrency and per-host pacing, and jittered exponential-backoff retries.
- **Security-conscious by design** — encrypted key vault, log redaction, report auto-escaping, SSRF redirection guards, and a documented threat model.
- **Quality-gated** — 756 passing tests at 89.54% branch coverage under a strict lint/type/test gate, with a reproducible benchmark baseline.
- **Reproducible distribution** — pinned dependencies, wheel/sdist, standalone binaries, Docker image, and checksummed releases via CI/CD.

---

72. Limitations and Future Enhancements

Table 72.1 – Known Limitations and Planned Enhancements

Area	Current limitation	Planned enhancement
Hash targets	No collector for file-hash (HASH) targets	Hash reputation module (key-gated)
IP scope	DNSBL limited to Spamhaus; IPv6 out of scope for DNSBL	Additional DNSBLs / IPv6 support
Port scanning	Active service/port identification intentionally omitted	Only if kept passive/authorised
Concurrency store	Semaphores are per-process	Distributed rate limiting for multi-worker deployments
API surface	No REST API	Optional self-hosted REST API
Collaboration	Single-user, local database	Optional self-hosted shared cases
AI	Remote provider is generic OpenAI-compatible	Provider-specific adapters and richer prompts
PDF	Depends on WeasyPrint system libraries	Bundled renderer / alternate backend
Anonymity	No traffic anonymisation	Optional Tor/proxy management
Version metadata	`core/config.py` default version string lags `core/constants.VERSION`	Unify version sourcing across modules

The most significant engineering limitation is the deliberate exclusion of any offensive capability — no exploitation, credential attacks, or authentication bypass — which is an ethical design constraint, not a technical gap. Several inconsistencies noted during authoring remain as documentation follow-ups: the Phase Plan summary still shows some completed phases unchecked, the README badge/status text lags the 1.0.0 release, and the default `app.version` in configuration is stale relative to the canonical constant.

---

73. Conclusion

IntelXtract demonstrates that a **local-first, account-free OSINT platform** can deliver the breadth of commercial tooling without its privacy, cost, and cloud dependencies. The system unifies 19 reconnaissance collectors, an asynchronous scan engine with bounded concurrency, an entity-centric correlation graph, a 22-rule deterministic risk model, and multi-format reporting behind both a command-line and a desktop interface.

Every phase was delivered against a uniform quality gate — formatting, linting, static typing, and testing on Python 3.13 and 3.14 — reaching 756 passing tests and 89.54% branch coverage, with a reproducible benchmark baseline and a fully automated release pipeline producing wheel/sdist, standalone binaries, a Docker image, and checksummed GitHub releases. Correctness and safety were treated as first-class: the risk engine is a pure function, the correlation graph is byte-stable, secrets are Fernet-encrypted and redacted from logs, report rendering is auto-escaped, and scanning is constrained to passive, authorised, publicly available sources.

The project therefore meets its objectives of breadth, determinism, extensibility, and privacy, while documenting its limits honestly. Its modular architecture and Plugin SDK leave clear room for the planned enhancements — REST API, additional key-gated providers, and richer collaboration — without compromising the local-first guarantees that define it.

---

74. References and Appendices

74.1 References

1. Python Software Foundation. *Python 3.13 / 3.14 Documentation.* https://docs.python.org/3/
2. aiohttp contributors. *aiohttp — Asynchronous HTTP Client/Server for asyncio.* https://docs.aiohttp.org/
3. Pallets. *Jinja2 Documentation.* https://jinja.palletsprojects.com/
4. Tiangolo, S. *Typer — CLI library for Python.* https://typer.tiangolo.com/
5. Rich contributors. *Rich — Text formatting for the terminal.* https://rich.readthedocs.io/
6. The Qt Company. *Qt 6 / PySide6 Documentation.* https://doc.qt.io/qtforpython/
7. Cryptography contributors. *cryptography — Fernet (symmetric encryption).* https://cryptography.io/
8. NetworkX developers. *NetworkX Documentation.* https://networkx.org/documentation/
9. aiosqlite contributors. *aiosqlite — asyncio interface to SQLite.* https://aiosqlite.omnilib.dev/
10. SQLite Consortium. *SQLite Documentation.* https://www.sqlite.org/docs.html
11. WeasyPrint contributors. *WeasyPrint — HTML to PDF.* https://weasyprint.org/
12. Haxx. *cURL / libcurl — transport reference (robots.txt, HTTP).* https://curl.se/
13. Balliet, S. *Have I Been Pwned API.* https://haveibeenpwned.com/API/v3
14. AbuseIPDB. *AbuseIPDB API Documentation.* https://docs.abuseipdb.com/
15. The Spamhaus Project. *Spamhaus DNSBL / ZEN.* https://www.spamhaus.org/
16. Internet Archive. *Wayback Machine / CDX API.* https://web.archive.org/
17. crt.sh / Sectigo. *Certificate Transparency log search.* https://crt.sh/
18. Chen, E., et al. *The Architecture of Open Source Applications* (asyncio concurrency patterns).
19. OWASP Foundation. *OWASP Secure Headers Project.* https://owasp.org/www-project-secure-headers/
20. RFC 7480–7484. *Registration Data Access Protocol (RDAP).* IETF.
21. RFC 7208. *Sender Policy Framework (SPF).* IETF.
22. RFC 7489. *Domain-based Message Authentication, Reporting, and Conformance (DMARC).* IETF.
23. RFC 6376. *DomainKeys Identified Mail (DKIM).* IETF.
24. `docs/architecture.md`, `docs/Database_Schema.md`, `docs/threat_model.md`, `docs/privacy.md`, `docs/performance.md` (IntelXtract project documentation).

74.2 Appendix A – Configuration Defaults

Key	Default
`app.name` / `app.version`	`IntelXtract` / `1.0.0` (`core.constants`)
`scan.default_mode` / `default_timeout`	`quick` / `30.0`
`scan.max_workers`	`8`
`cache.enabled` / `ttl` / `max_size`	`true` / `300.0` / `1024`
`logging.level` / `file`	`INFO` / `logs/intelxtract.log`
`logging.max_bytes` / `backup_count`	`1048576` / `3`
`network.user_agent`	`IntelXtract/...`
Env prefix / config path	`INTELXTRACT_` / `.env`, `config/settings.json`

74.3 Appendix B – Correlation Edge Types

Edge	Meaning
`resolves_to`	domain → IP / alias
`registered_by`	domain → registrar organisation
`issued_to`	domain / certificate → issuer
`same_email`	email → domain / username
`part_of`	base domain → subdomain
`mentions`	publisher domain → target
`profile`	username → social profile
`uses`	domain → technology

74.4 Appendix C – Screenshot Placeholders

The following figures should be populated with screenshots before submission:

Figure	Screenshot
Figure 56.2	Dashboard page with a completed scan
Figure 56.3	Quick Scan page configuration
Figure 56.4	Results page with severity filters
Figure 56.5	Correlation graph visualisation
Figure 56.6	Cases page with notes and evidence
Figure 56.7	Reports page with format selection
Figure 56.8	API Manager page (masked keys)
Figure 56.9	Scheduler page

74.5 Appendix D – Reproduction Commands

```
# Quality gate
export PATH="$PWD/.venv/bin:$PATH" && black . && ruff check . && mypy . && pytest -p no:cachep

# Benchmark
python scripts/bench.py --json

# Run a scan and export every format
intelxtract scan example.com --mode deep
intelxtract scan example.com --mode deep --format json
intelxtract report 1 --format html --out exports/report.html
```

End of report.